

Response to Reviewers

Dear Editor,

We sincerely thank the editor and reviewers for their careful reading of our manuscript and for the constructive comments. We have revised the manuscript, Supplementary Information, figures, tables, equations, references, and code/data availability statement to make the data-generation workflow, optimizer formulation, comparison protocol, physical stress-test boundary, and climate-sensitivity boundary explicit.

This response letter has been reorganized as a comment-level evidence package. Each reviewer comment is reproduced under its own comment heading. The response then states the revision, identifies where the manuscript was changed, and displays the relevant revised manuscript, Supplementary Information, figure, table, equation, or local reference material directly under that comment.

Manuscript title: Surrogate-conditioned optimizer ranking in block-scale urban energy design: A comparison of policy learning and Pareto search

Manuscript ID: APEN-D-26-08148

Sincerely,
The Authors

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Summary of major revisions

The revision reframes the paper as a surrogate-conditioned, representation-aware comparison for block-scale urban energy design. The manuscript now distinguishes generated feasible block morphologies, 12 morphology descriptors, analytic response generation, DNN surrogate prediction, descriptor-space optimization, nearest-feasible-block projection, physics-based cross-model stress testing, and cross-climate sensitivity analysis. This revision keeps the interpretation boundary visible: the results support an early-stage design-support workflow and do not claim universal optimizer superiority or physical ranking transfer.

Table R1: Editor-facing overview of major revision packages.

Revision package	Main comments addressed	Main manuscript changes	Supplementary support	Information
Manuscript reframing	R1-1, R1-15, R4-1, R4-5	Abstract, Introduction, Discussion, and Conclusions rewritten around surrogate-conditioned design support	SI S1–S6 aligned with the revised scope	
Analytic response generation	R2-1, R2-7, R3-5	Section 2.2 and Table 2 define analytic target assignment	SI S1.4 gives equations, stochastic terms, and clipping ranges	
Sampling and descriptor coverage	R1-5, R2-4	Section 2.1 and Table 1 define generated feasible morphologies and 12 descriptors	SI S1.2–S1.5 gives prototype settings, dependencies, PCA, and descriptor distributions	
Surrogate assessment	R1-7, R1-8, R2-3	Section 2.3, Table 3, and Figs. 4–5 report repeated and boundary-oriented checks	SI S2 gives scale study, residuals, OSLI holdout, outer-shell, and tail holdouts	
DDPG formulation and reward	R1-10, R1-11, R1-13, R2-5, R3-3	Section 2.4 and Figs. 2–3 define serialized static black-box search and corrected reward	SI S3.1 gives seed-level behaviour	
Benchmark fairness and projection	R3-2, R4-4	Tables 4–5 and Figs. 7–8 separate retained outputs, equal-size metrics, and feasible projection	SI S3 and S5 provide secondary metrics and complete projection notes	
Physical and climate boundaries	R1-14, R2-2, R2-3, R4-2	Section 2.6, Section 3.5, Fig. 9, Fig. 10, and Table 6 define stress-test and climate scope	SI S5–S6 give direct stress-test table, EG proxy note, and climate metadata	
Readability, figures, units, and reproducibility	R1-2, R1-3, R1-6, R1-12, R2-8, R3-1, R3-4	Abbreviations, Tables 1–2, Figs. 1–10, Data and code availability, references, and captions revised	SI figures and tables follow the same unit and terminology conventions	

The Supplementary Information was reorganized to provide the equations, validation protocols, optimizer analyses, physical case results, and climate case results that support the revised manuscript. Table R1 is only an editor-facing overview; the evidence for each reviewer comment appears under the corresponding comment below.

► **Reviewer 1.**

✓ We thank Reviewer 1 for the detailed comments on readability, terminology, figures, software context, parameter selection, validation, cross-validation, framework structure, DDPG formulation, units, case generality, and broader-audience accessibility. The revised manuscript responds by simplifying the framing, adding definitions and tables, expanding the method explanation, and keeping the claims within the analytic response-generation and limited sensitivity-analysis scope.

■ **Comment R1-1.**

In the Abstract, the paragraph is too long (more than 250 words), which can be simplified. Furthermore, it is recommended that authors point out the significance of the work with more brief but powerful expressions.

Response.

✓ We rewrote the Abstract to state the study significance more directly. The revised Abstract emphasizes block-scale energy-design support, the shared surrogate environment, feasible-block projection, the 24-case cross-model stress test, and the four-block cross-climate sensitivity analysis. The wording avoids presenting the revision as a universal optimizer-superiority claim.

Revisions made in the manuscript.

The relevant revisions appear in the Abstract, the final Introduction contribution paragraph, and the Conclusions. The abstract now reports the repeated-validation values, projection compression, stress-test boundary, and climate boundary in one bounded narrative.

Relevant revised manuscript and supporting evidence.

Main manuscript, Abstract.

Block-scale urban energy design requires optimization methods that connect urban form with building energy demand, rooftop photovoltaic potential, and winter solar access during early-stage planning. This study develops a surrogate-conditioned framework to compare policy learning and Pareto search for residential-block morphology optimization. Feasible block layouts are generated first and converted into 12 morphology descriptors. A morphology-to-performance response surface then assigns three targets to each sample: Energy Use Intensity, Energy Generation, and winter sunlight hours. These targets are denoted EUIt, EG, and H throughout the paper. The 2000-sample DNN surrogate selected from nested 500/1000/1500/2000-sample pools provides the common environment for DDPG and NSGA-II. Repeated five-fold validation gives mean nMAE values of 0.0176, 0.0265, and 0.0145 for EUIt, EG, and H. Under the shared surrogate and matched query budget, NSGA-II attains near-ceiling descriptor-space HV, while DDPG shows preference-dependent performance. Projection to generated feasible blocks

compresses the 2000-candidate NSGA-II archive to 51 unique feasible blocks, indicating that descriptor-space coverage and feasible-morphology diversity differ substantially. A 24-case cross-model stress test completes all locked cases and shows limited metric and rank consistency between analytic and physics-based outputs. A four-block cross-climate sensitivity analysis shows stable EG ordering but climate-dependent EUIt and H responses. The study contributes a representation-aware urban design workflow that links surrogate search, feasible-block projection, and sensitivity analysis to improve the interpretability of optimizer comparisons for early-stage urban energy planning.

Scope boundary.

The Abstract states that the work supports a representation-aware design workflow; it does not claim direct physical optimizer superiority.

■ **Comment R1-2.**

Overall, there are too many acronyms which decreases the paper's readability. Moreover, symbols should be added.

Response.

✓ We added an Abbreviations block and clarified the major symbols used for energy targets, learning methods, optimizer metrics, and morphology descriptors. We also revised Table 1 so that the 12 descriptors are presented with units, construction notes, and interpretation.

Revisions made in the manuscript.

The relevant revisions appear in the Abbreviations block, Table 1, Section 2.1, and Supplementary Section S1.3.

Relevant revised manuscript and supporting evidence.

Main manuscript, Abbreviations block excerpt.

EUIt is defined as Energy Use Intensity, EG as Energy Generation, H as winter sunlight hours, DNN as deep neural network, DDPG as Deep Deterministic Policy Gradient, NSGA-II as Non-dominated Sorting Genetic Algorithm II, HV as Hypervolume, IGD as Inverted Generational Distance, nMAE as normalized mean absolute error, and OSLI as open-space location index.

Main manuscript, Table 1 excerpt.

Descriptor	Unit or type	Construction	Interpretation
FAR	–	$A_{\text{gross}}/A_{\text{block}}$	density
SD	m	$h_{\text{max}} - \bar{h}$	height variation
AF	floors	$A_{\text{gross}}/A_{\text{foot}}$	linked to FAR, BD
AR_{ew}	–	$\bar{h}/W_{ew}$	east-west canyon
AR_{ns}	–	$\bar{h}/W_{ns}$	north-south canyon
SVF	–	analytic openness proxy	sky-view openness
BD	–	$A_{\text{foot}}/A_{\text{block}}$	footprint density
OSR	–	$A_{\text{open}}/A_{\text{gross}}$	open-space ratio
SC	–	$A_{\text{env}}/A_{\text{gross}}$	envelope exposure
PAR	m^{-1}	$P_{\text{sum}}/A_{\text{foot}}$	compactness
θ	deg	sampled in $[-45, 45]$	orientation
OSLI	integer	0–8	open-space cell

Revised Table 1 excerpt. The 12 quantities are derived morphology descriptors, not independent design variables.

Supplementary Material, Section S1.3 paragraph.

The descriptor set is FAR, SD, AF, AR_{ew} , AR_{ns} , SVF, BD, OSR, SC, PAR, theta, and OSLI. FAR is numerically linked to BD and AF, while OSR is linked to footprint area and gross floor area. OSLI is integer-valued by construction. In the locked descriptor-dependency check, six principal components explain 95% of descriptor variance, and exact duplicate descriptor rows are absent in the 2000-row canonical pool.

Scope boundary.

The symbols describe derived morphology descriptors and target quantities; they do not redefine the descriptors as independent design variables.

■ **Comment R1-3.**

Figure 1: The caption should be improved to explain the content of the figure in more detail.

Response.

✓ We revised Figure 1 and its caption so that the workflow explicitly separates feasible morphology generation, analytic response assignment, surrogate-assisted optimization, feasible-block projection, and sensitivity analysis.

Revisions made in the manuscript.

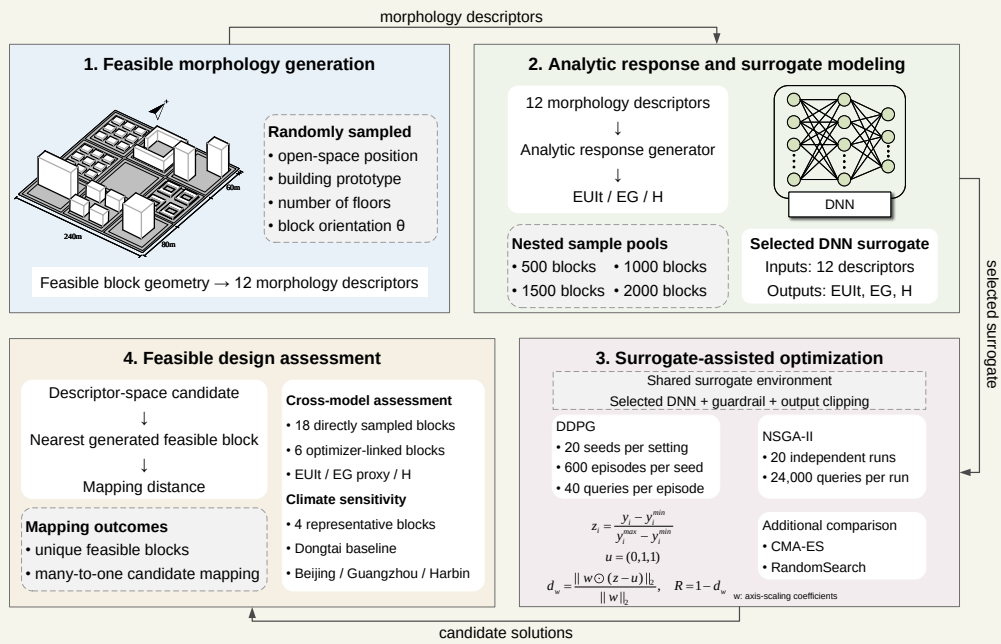
The relevant revisions appear in Section 2 and Fig. 1. The caption now describes the figure as a workflow and avoids implying that all 2000 samples were direct annual physical simulations.

Relevant revised manuscript and supporting evidence.

Main manuscript, methodology transition before Fig. 1.

The methodology links morphology generation, analytic target assignment, surrogate-assisted optimization, and post-optimization assessment within one block-scale design workflow. The workflow distinguishes three representations: generated feasible block geometries, 12 morphology descriptors used by the surrogate and optimizers, and locked block cases examined through physics-based and climate-sensitive calculations. This distinction is essential because the optimizer operates in descriptor space, whereas design interpretation ultimately depends on generated block geometries. Figure 1 summarizes the computational sequence.

Main manuscript, Fig. 1.



Revised Fig. 1. Workflow linking feasible morphology generation, analytic response assignment, surrogate-assisted optimization, feasible-block projection, and sensitivity analysis.

Scope boundary.

The 2000-sample pool is assigned analytic responses; selected locked block cases are examined later by physics-based and climate-sensitive calculations.

■ Comment R1-4.

Line 36-38: According to the Chinesees requires reference. Moreover, the authors need to specify some references for the Grasshopper platform. Please describe the general context of the softwares (tools) and why that softwares are suitable for your analysis.

Response.

✓ We revised the Introduction to connect the work to Chinese urban districts, urban morphology, rooftop photovoltaic potential, and climate-resilient planning. We also clarified the tool and workflow context by separating analytic response generation from selected physics-based cross-model calculations.

Revisions made in the manuscript.

The relevant revisions appear in Sections 1.1, 1.2, 2.2, and 2.6. The revision cites urban morphology, surrogate modelling, simulation-based optimization, and climate-resilient urban-energy studies. It also clarifies that the physics-based workflow is used for selected locked cases, not for direct simulation of the full 2000-sample pool.

Relevant revised manuscript and supporting evidence.

Main manuscript, Motivation paragraph.

Block morphology influences energy performance through coupled physical mechanisms. Density and height alter heating and cooling demand by changing envelope exposure, mutual shading, and micro-climatic context. Solar access and daylight availability depend on roof exposure, sky-view conditions, block orientation, and open-space arrangement. Recent large-scale studies further show that urban spatial structure can shape building energy demand across broad urban areas and under extreme climate events, which makes morphology-sensitive design relevant beyond isolated building cases.

Main manuscript, Section 2.2 paragraph.

This module provides a controlled morphology-to-performance response surface for optimizer comparison. The nested 500/1000/1500/2000-sample pools are deterministic subsets of the same generated morphology pool under the recorded random seed and morphology-generation protocol. The physics-based case calculations are conducted subsequently on locked generated blocks to assess how the analytic target values relate to EnergyPlus- and Radiance-component outputs in selected cases.

Supplementary Material, Section S5.3 paragraph.

The 18 direct feasible cases completed both EnergyPlus and Radiance-component runs. EUI agreement is weak, EG is represented by a GHI-based rooftop-PV proxy, and H is represented by January 20 windowsill direct-sun hours. The nMAE and rank-correlation values therefore define a stress-test boundary for the tested direct cases.

Scope boundary.

The software-supported physical calculations apply to selected locked cases. The full benchmark target assignment remains analytic response generation.

Local references.

Javanroodi, K., Perera, A. T. D., et al. (2023). Designing climate resilient energy systems in complex urban areas considering urban

morphology: A technical review. *Advances in Applied Energy*, 12. <https://doi.org/10.1016/j.adapen.2023.100155>

Du, L., Wang, H., et al. (2025). Impact of block form on building energy consumption, urban microclimate and solar potential: A case study of Wuhan, China. *Energy and Buildings*, 328. <https://doi.org/10.1016/j.enbuild.2024.115224>

Wang, H., Wu, Z., Gu, W., et al. (2026). Towards climate-resilient cities: Assessing and planning urban energy demand under extreme climate events. *Energy and Buildings*, 337, 117436. <https://doi.org/10.1016/j.enbuild.2026.117436>

■ Comment R1-5.

Line 49: It would be good to describe why you select 12 parameters for building models.

Response.

✓ We clarified that the 12 inputs are morphology descriptors calculated after feasible block generation. They were selected to summarize density, height variation, canyon aspect ratio, sky-view openness, footprint density, open-space ratio, compactness, orientation, and open-space position.

Revisions made in the manuscript.

The relevant revisions appear in Section 2.1, Table 1, Supplementary Sections S1.2–S1.5, Table S2, and Fig. S1.

Relevant revised manuscript and supporting evidence.

Main manuscript, Section 2.1 paragraphs.

The morphology generator first constructs feasible block layouts and only then calculates descriptor values. Each block is represented as a 240 m × 240 m square divided into a 3 × 3 grid of 80 m × 80 m land units. One cell is assigned as open space, and the remaining eight cells are populated with residential prototypes. For each generated block, the generator samples the open-space position, building prototype, number of floors, and orientation angle. The orientation angle is sampled within -45 deg to 45 deg. The resulting block is therefore a generated feasible morphology, while the 12 input quantities used by the surrogate are descriptors derived from that morphology.

The descriptor set in Table 1 includes FAR, SD, AF, AR_{ew}, AR_{ns}, SVF, BD, OSR, SC, PAR, theta, and OSLI. These quantities are derived morphology descriptors with geometric dependencies. For example, FAR, BD, AF, and OSR are linked through the same footprint and gross-floor-area quantities, while OSLI is an integer descriptor of open-space position. Consequently, optimizer outputs are first interpreted in descriptor space and then mapped to generated feasible blocks before morphology-level conclusions are drawn.

Main manuscript, Table 1 excerpt.

Descriptor	Unit or type	Construction	Interpretation
FAR	–	$A_{\text{gross}}/A_{\text{block}}$	density
SD	m	$h_{\text{max}} - \bar{h}$	height variation
AF	floors	$A_{\text{gross}}/A_{\text{foot}}$	linked to FAR, BD
AR_{ew}	–	$\bar{h}/W_{ew}$	east-west canyon
AR_{ns}	–	$\bar{h}/W_{ns}$	north-south canyon
SVF	–	analytic openness proxy	sky-view openness
BD	–	$A_{\text{foot}}/A_{\text{block}}$	footprint density
OSR	–	$A_{\text{open}}/A_{\text{gross}}$	open-space ratio
SC	–	$A_{\text{env}}/A_{\text{gross}}$	envelope exposure
PAR	m^{-1}	$P_{\text{sum}}/A_{\text{foot}}$	compactness
θ	deg	sampled in $[-45, 45]$	orientation
OSLI	integer	0–8	open-space cell

Revised Table 1 excerpt. The 12 quantities are derived morphology descriptors, not independent design variables.

Supplementary Material, Table S2 excerpt.

Check	Value	Interpretation
FAR minus BD times AF	1.33×10^{-15}	numerical roundoff
OSR relation residual	2.22×10^{-16}	footprint relation holds
OSLI integer consistency	1.0	all OSLI values are integer
Exact duplicate count	0	no exact duplicate descriptor rows
PCA components for 95% variance	6	descriptor space is structured

Supplementary Material, Fig. S1.

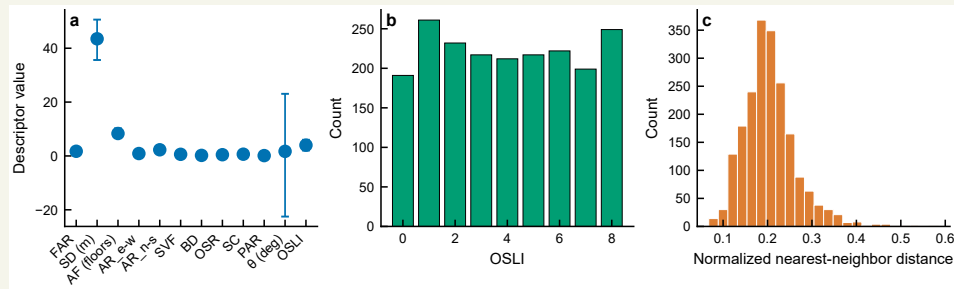


Fig. S1 excerpt. Descriptor distributions and nearest-neighbour distances in the generated morphology pool.

Scope boundary.

The 12 inputs are derived morphology descriptors, not independent building-model variables.

■ **Comment R1-6.**

Table 1: The author could add an additional column on the right to describe Factors (Floor area ratio, Setback, diversity, etc.). Table 2: The parameters are not assumptions; it seems the baseline or operating parameters for the building energy models.

Response.

✓ We revised Table 1 into a descriptor table with construction and interpretation columns. We also reframed Table 2 as an evaluation-mode table instead of an assumption table.

Revisions made in the manuscript.

The relevant revisions appear in Tables 1 and 2 and in Sections 2.1–2.2.

Relevant revised manuscript and supporting evidence.

Main manuscript, Table 1 excerpt.

Descriptor	Unit or type	Construction	Interpretation
FAR	–	$A_{\text{gross}}/A_{\text{block}}$	density
SD	m	$h_{\text{max}} - \bar{h}$	height variation
AF	floors	$A_{\text{gross}}/A_{\text{foot}}$	linked to FAR, BD
AR_{ew}	–	$\bar{h}/W_{ew}$	east-west canyon
AR_{ns}	–	$\bar{h}/W_{ns}$	north-south canyon
SVF	–	analytic openness proxy	sky-view openness
BD	–	$A_{\text{foot}}/A_{\text{block}}$	footprint density
OSR	–	$A_{\text{open}}/A_{\text{gross}}$	open-space ratio
SC	–	$A_{\text{env}}/A_{\text{gross}}$	envelope exposure
PAR	m^{-1}	$P_{\text{sum}}/A_{\text{foot}}$	compactness
θ	deg	sampled in $[-45, 45]$	orientation
OSLI	integer	0–8	open-space cell

Revised Table 1 excerpt. The 12 quantities are derived morphology descriptors, not independent design variables.

Main manuscript, Table 2 excerpt.

Mode	Source	Output	Boundary
Analytic response generation	generated descriptors	EUI, EG, H	analytic surface
Surrogate prediction	12 descriptors	DNN outputs	selected checkpoint
Feasible projection	descriptor candidate	nearest block	nearest neighbour
Physical stress test	locked blocks	physical outputs	24 cases
Climate sensitivity	four blocks	weather deltas	no retraining

Revised Table 2 excerpt. Evaluation modes are separated by source, output, and interpretation boundary.

Scope boundary.

Table 1 defines morphology descriptors. Table 2 defines analysis modes and their boundaries; it is no longer presented as a building-energy assumption table.

■ **Comment R1-7.**

The author may need to justify the methodology used. I think it would be great to have model validation studies using published literature data or exp. data. Moreover, the authors need to describe the underlying assumptions of their models.

Response.

✓ We expanded the methodology and validation-scope discussion. The manuscript now states the analytic response-generation assumptions, reports repeated and boundary-oriented surrogate assessments, and adds a limited physics-based cross-model stress test. The stress test is interpreted conservatively because agreement with physics-based outputs is limited.

Revisions made in the manuscript.

The relevant revisions appear in Sections 2.2, 2.3, 2.6, 3.1, 3.5, 4.2, Table 3, Table 6, Fig. 5, Fig. 9, and Supplementary Sections S1, S2, and S5.

Relevant revised manuscript and supporting evidence.

Main manuscript, Section 2.2 paragraph.

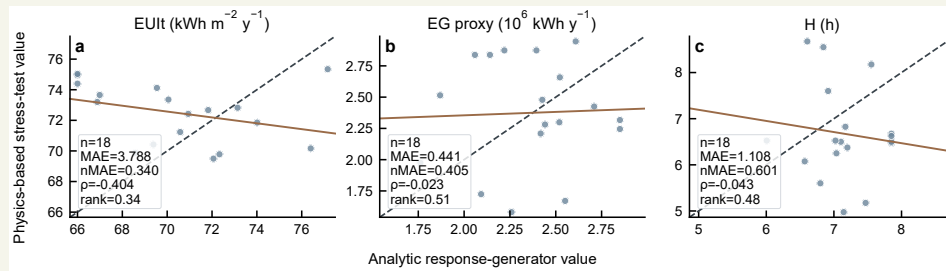
In this study, the analytic response generator is the target-assignment module that maps a generated morphology descriptor vector, denoted $\mathbf{x}$, to three target responses, denoted $\mathbf{y}^{ana} = [\text{EUIt}, \text{EG}, H]^T$. The mapping uses explicit formulas that combine density, openness, aspect ratio, sky-view factor, orientation, and open-space position to represent demand, rooftop-generation, and sunlight-access responses. A small stochastic term is added during dataset construction, and the target values are then limited to the response ranges used for surrogate training.

Main manuscript, Table 3 excerpt.

Validation regime	EUIt nMAE	EG nMAE	H nMAE	EUIt ρ	EG ρ	H ρ
Repeated 5×5 CV	0.0176	0.0265	0.0145	0.991	0.984	0.995
Leave-one-OSLI-out	0.0185	0.0342	0.0185	0.990	0.972	0.992
Outer-shell holdout	0.0278	0.0335	0.0208	0.985	0.975	0.994
Feature-tail holdout	0.0230	0.0287	0.0198	0.986	0.983	0.992

Revised Table 3 excerpt. nMAE and Spearman ρ compare DNN predictions with analytic target values.

Main manuscript, Fig. 9.



Revised Fig. 9. Cross-model stress test for 18 direct generated feasible blocks.

Supplementary Material, Table S4 excerpt.

Target	Count	MAE	nMAE	Spearman ρ
EUIt	18	3.788	0.340	-0.404
EG GHI proxy	18	0.441	0.405	-0.023
H	18	1.108	0.601	-0.043

Revised Table 6 and Table S4 excerpt. The direct feasible stress-test cases show limited metric and rank consistency.

Scope boundary.

The physics-based cross-model stress test defines an interpretation boundary. It does not validate optimizer ranking transfer.

Comment R1-8.

The author may need to describe what the five-fold cross validation means.

Response.

✓ We added an explanation of the repeated five-fold validation protocol and reported the target-wise metrics directly in the manuscript and Supplementary Information.

Revisions made in the manuscript.

The relevant revisions appear in Section 2.3, Table 3, Fig. 5, and Supplementary Sections S2.2–S2.5.

Relevant revised manuscript and supporting evidence.

Main manuscript, Section 2.3 paragraph.

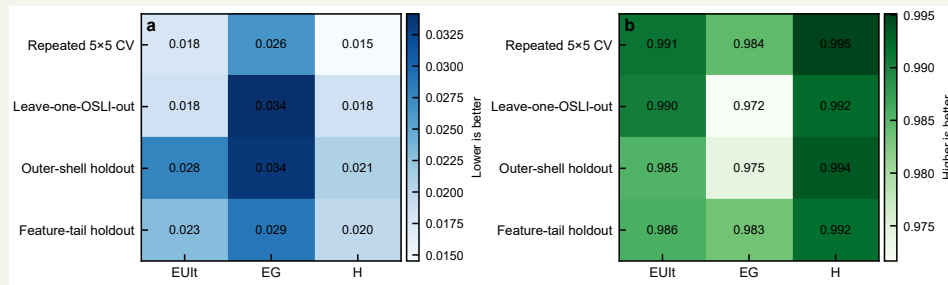
The selected surrogate is assessed through repeated five-fold cross-validation, leave-one-OSLI-out validation, outer-shell holdout, and feature-tail holdout. The repeated five-fold protocol evaluates interpolation performance under randomized folds. The leave-one-OSLI-out protocol tests whether the surrogate remains stable when one open-space-position class is withheld. The outer-shell and feature-tail holdouts examine surrogate behaviour near the boundary of the sampled descriptor domain.

Main manuscript, Table 3 excerpt.

Validation regime	EUI t nMAE	EG nMAE	H nMAE	EUI t ρ	EG ρ	H ρ
Repeated 5×5 CV	0.0176	0.0265	0.0145	0.991	0.984	0.995
Leave-one-OSLI-out	0.0185	0.0342	0.0185	0.990	0.972	0.992
Outer-shell holdout	0.0278	0.0335	0.0208	0.985	0.975	0.994
Feature-tail holdout	0.0230	0.0287	0.0198	0.986	0.983	0.992

Revised Table 3 excerpt. nMAE and Spearman ρ compare DNN predictions with analytic target values.

Main manuscript, Fig. 5.



Revised Fig. 5. Surrogate assessment across random, grouped, and boundary-oriented validation regimes.

Supplementary Material, Section S2.2 paragraph.

Repeated five-fold cross-validation gives mean nMAE values of 0.0176 for EUI t, 0.0265 for EG, and 0.0145 for H. The corresponding Spearman correlations are 0.991, 0.984, and 0.995. These values support low error within the analytic response-generation regime.

Scope boundary.

The validation assesses surrogate prediction against analytic target values, not direct physical truth.

■ Comment R1-9.

The framework has two components: The first The second ...

Response.

✓ We revised the framework description so that the components are no longer implicit. The workflow is now presented as morphology generation and analytic target assignment, surrogate optimization, feasible projection, and selected sensitivity analyses.

Revisions made in the manuscript.

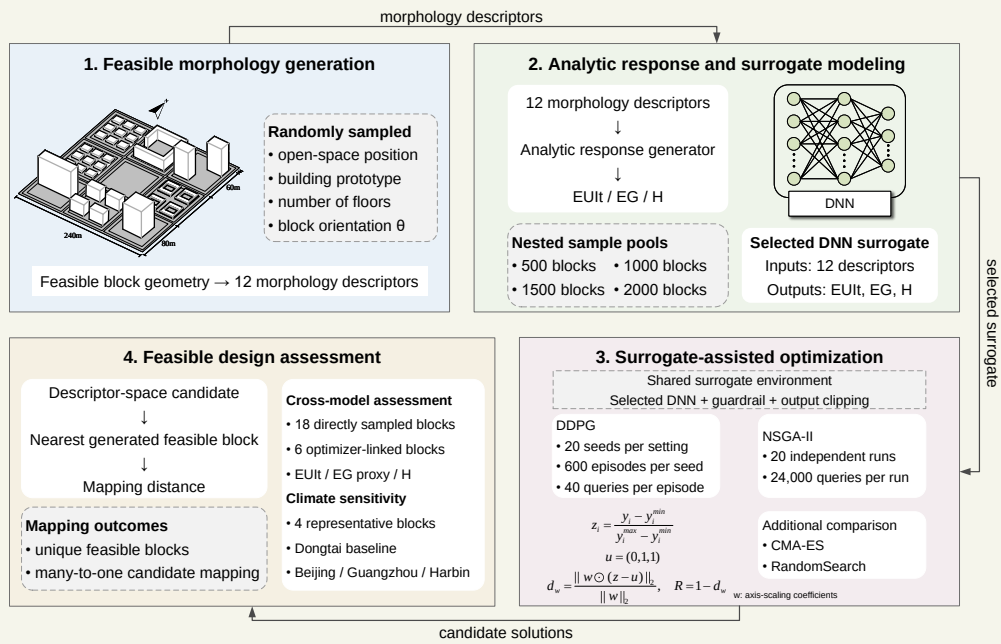
The relevant revisions appear in the Methodology opening, Fig. 1, Table 2, and Section 2.6.

Relevant revised manuscript and supporting evidence.

Main manuscript, methodology opening paragraph.

The methodology links morphology generation, analytic target assignment, surrogate-assisted optimization, and post-optimization assessment within one block-scale design workflow. The workflow distinguishes three representations: generated feasible block geometries, 12 morphology descriptors used by the surrogate and optimizers, and locked block cases examined through physics-based and climate-sensitive calculations.

Main manuscript, Fig. 1.



Revised Fig. 1. Workflow linking feasible morphology generation, analytic response assignment, surrogate-assisted optimization, feasible-block projection, and sensitivity analysis.

Main manuscript, Table 2 excerpt.

Mode	Source	Output	Boundary
Analytic response generation	generated descriptors	EUI, EG, H	analytic surface
Surrogate prediction	12 descriptors	DNN outputs	selected checkpoint
Feasible projection	descriptor candidate	nearest block	nearest neighbour
Physical stress test	locked blocks	physical outputs	24 cases
Climate sensitivity	four blocks	weather deltas	no retraining

Revised Table 2 excerpt. Evaluation modes are separated by source, output, and interpretation boundary.

Scope boundary.

Fast surrogate search and post-optimization feasible/physical/climate assessment are separate work-flow stages.

■ **Comment R1-10.**

Please add a reference for Markov Decision Process.

Response.

✓ We revised the RL framing and added local references for foundational reinforcement learning and DDPG. The manuscript no longer presents the task as a natural physical-time MDP. Instead, DDPG is described as serialized static black-box search on a surrogate response surface.

Revisions made in the manuscript.

The relevant revisions appear in Sections 1.2 and 2.4. The reward and DDPG architecture are shown in Section 2.4 and Fig. 3.

Relevant revised manuscript and supporting evidence.

Main manuscript, Section 1.2 paragraph.

Deep reinforcement learning provides another family of search methods, but its role in morphology design requires careful framing. General DRL reviews and foundational DQN and DDPG studies establish policy learning for sequential decision-making and continuous control. In the present study, DDPG is adopted as a continuous-action policy-learning comparator because it provides a clear actor-critic mechanism for generating descriptor queries in a bounded surrogate environment.

Main manuscript, Section 2.4 paragraph.

The DDPG environment serializes repeated queries to a static guarded surrogate. At the beginning of each episode, a random descriptor query is evaluated to construct the initial normalized three-target state. At each subsequent step, the actor outputs an absolute 12-dimensional normalized descriptor action. This action is not an incremental modification of the preceding block. The action is denormalized to descriptor bounds, evaluated by the guarded surrogate, and converted to the next normalized target state and scalar reward. The 40-step horizon therefore functions as a fixed sequence of policy-training queries on a static surrogate response surface.

Main manuscript, reward formula excerpt.

$$z_i = \frac{y_i - y_i^{\min}}{y_i^{\max} - y_i^{\min}}.$$
$$\mathbf{u} = [0, 1, 1]^T.$$

$$d_w = \frac{\|\mathbf{w} \odot (\mathbf{z} - \mathbf{u})\|_2}{\|\mathbf{w}\|_2}, \quad R = 1 - d_w.$$

Scope boundary.

DDPG is used here as a serialized static black-box search comparator, not as physical-time morphology control.

Local references.

- Wang, X., Wang, S., et al. (2024). Deep reinforcement learning: A survey. *IEEE Transactions on Neural Networks and Learning Systems*, 35(4), 5064–5078. <https://doi.org/10.1109/tnnls.2022.3207346>
- Mnih, V., Kavukcuoglu, K., et al. (2015). Human-level control through deep reinforcement learning. *Nature*, 518(7540), 529–533. <https://doi.org/10.1038/nature14236>
- Lillicrap, T. P., Hunt, J. J., et al. (2015). Continuous control with deep reinforcement learning. *arXiv preprint arXiv:1509.02971*. <https://doi.org/10.48550/arXiv.1509.02971>

■ Comment R1-11.

Figure 3: The manuscript does not include the explanation of the Figure 3. Moreover, the authors need to explain the symbols and equations in the Figure 3.

Response.

✓ We revised the DDPG section and Fig. 3 caption. The manuscript now explains state, action, reward, online actor, online critic, target networks, replay buffer, and the reward equations.

Revisions made in the manuscript.

The relevant revisions appear in Section 2.4 and Fig. 3.

Relevant revised manuscript and supporting evidence.

Main manuscript, Section 2.4 paragraph defining state/action/episode.

The DDPG environment serializes repeated queries to a static guarded surrogate. At the beginning of each episode, a random descriptor query is evaluated to construct the initial normalized three-target state. At each subsequent step, the actor outputs an absolute 12-dimensional normalized descriptor action. This action is not an incremental modification of the preceding block. The action is denormalized to descriptor bounds, evaluated by the guarded surrogate, and converted to the next normalized target state and scalar reward.

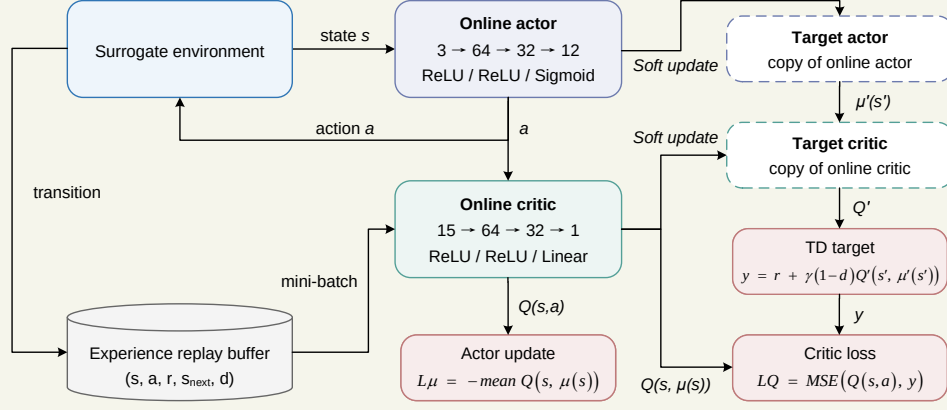
Main manuscript, reward equations.

$$z_i = \frac{y_i - y_i^{\min}}{y_i^{\max} - y_i^{\min}}.$$

$$\mathbf{u} = [0, 1, 1]^\top.$$

$$d_w = \frac{\|\mathbf{w} \odot (\mathbf{z} - \mathbf{u})\|_2}{\|\mathbf{w}\|_2}, \quad R = 1 - d_w.$$

Main manuscript, Fig. 3.



Revised Fig. 3. DDPG actor-critic architecture and target-network update structure used in the surrogate-query environment.

Scope boundary.

The actor output is an absolute descriptor action; it is not incremental physical morphology editing.

■ Comment R1-12.

Figure 4: For parity plots, the ticks and limits of the x- and y-axes should be same. All plots should be rescaled. Please specify unit in each axis.

Response.

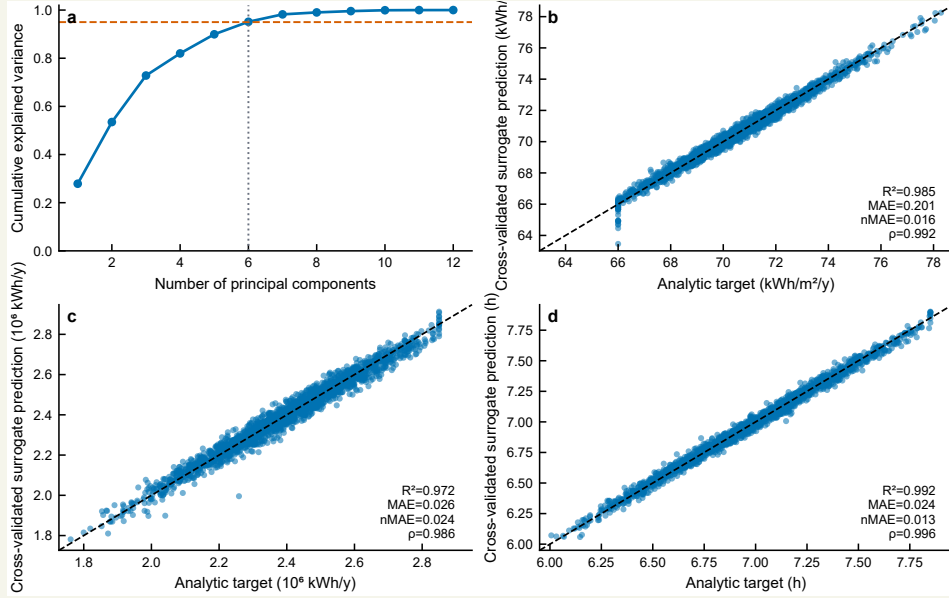
✓ We regenerated the parity and validation figures with matched parity axes and explicit units where applicable. The revised figures and tables consistently use EUI in $\text{kWh m}^{-2} \text{y}^{-1}$, EG in 10^6kWh y^{-1} , and H in hours.

Revisions made in the manuscript.

The relevant revisions appear in Fig. 4, Fig. 5, Fig. 9, Table 3, Table 6, and the Supplementary Information captions and tables.

Relevant revised manuscript and supporting evidence.

Main manuscript, Fig. 4.



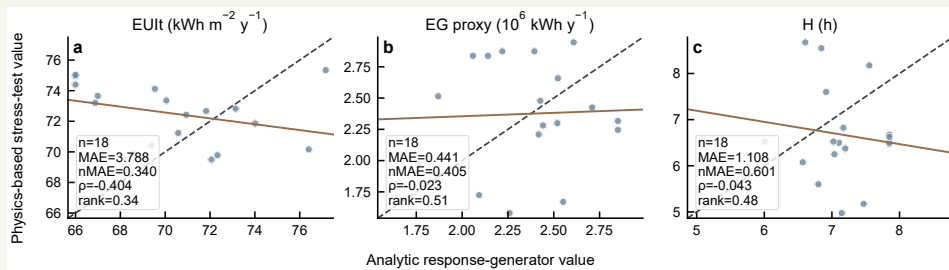
Revised Fig. 4. Descriptor coverage and cross-validated surrogate predictions.

Main manuscript, Table 3 excerpt with units in target definitions.

Validation regime	EUI _t nMAE	EG nMAE	H nMAE	EUI _t ρ	EG ρ	H ρ
Repeated 5×5 CV	0.0176	0.0265	0.0145	0.991	0.984	0.995
Leave-one-OSLI-out	0.0185	0.0342	0.0185	0.990	0.972	0.992
Outer-shell holdout	0.0278	0.0335	0.0208	0.985	0.975	0.994
Feature-tail holdout	0.0230	0.0287	0.0198	0.986	0.983	0.992

Revised Table 3 excerpt. nMAE and Spearman ρ compare DNN predictions with analytic target values.

Main manuscript, Fig. 9.



Revised Fig. 9. Cross-model stress test for 18 direct generated feasible blocks.

Scope boundary.

These unit and axis revisions improve traceability; they do not change the underlying numerical results.

■ **Comment R1-13.**

Figure 6: Please explain what the episode means in the plots.

Response.

✓ We defined an episode as a fixed 40-query training horizon in the serialized DDPG surrogate environment. The revised text states that an episode is not physical time evolution of one block.

Revisions made in the manuscript.

The relevant revisions appear in Section 2.4, Section 3.2, Fig. 2, Fig. 6, and Supplementary Section S3.1.

Relevant revised manuscript and supporting evidence.

Main manuscript, Section 2.4 paragraph.

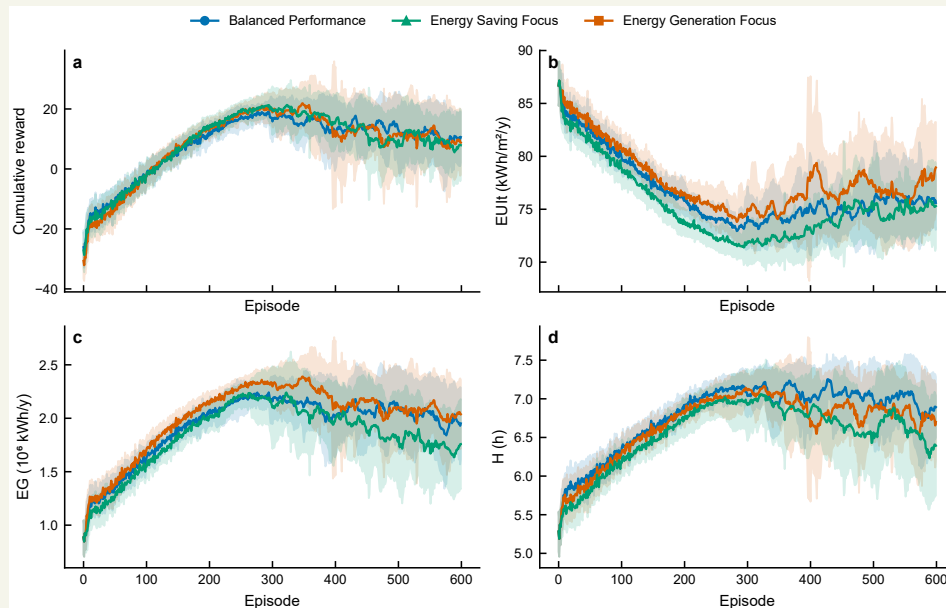
The 40-step horizon therefore functions as a fixed sequence of policy-training queries on a static surrogate response surface.

Main manuscript, Section 3.2 paragraphs.

Figure 6 describes how the DDPG agents interact with the guarded surrogate during training. Across the three preference scenarios, rewards increase rapidly during early training and then enter a high-reward region with visible seed-to-seed variability. Because each episode contains a fixed sequence of 40 surrogate queries, the trajectories represent policy-training behaviour on a static response surface. They should therefore be read as search trajectories over descriptor queries, not as the physical evolution of one block design.

The later portions of the curves remain informative. The agents can repeatedly reach favourable surrogate regions, yet the best-to-final gap indicates that late-stage regression remains present. This behaviour is consistent with the retained-output strategy used in Table 4, where one best scalarized candidate is retained from each seed.

Main manuscript, Fig. 6.



Revised Fig. 6. DDPG training trajectories across the three preference scenarios.

Supplementary Material, Section S3.1 paragraph.

Each DDPG scenario uses 20 seeds, 600 episodes, and 40 surrogate interactions per episode. The seed-level summaries record best reward, final reward, plateau episode, and best-to-final regression. These quantities describe policy-training behaviour under serialized surrogate queries on a static surrogate response surface.

Scope boundary.

An episode is a surrogate-query sequence, not physical time.

Comment R1-14.

It would be better to conduct more generalized case studies, including different locations (different weather conditions in China) or different building types (an energy-efficient building, an old building, etc.).

Response.

✓ We added a limited cross-climate sensitivity analysis using the same four generated blocks under Harbin, Beijing, and Guangzhou weather. We did not add new building types because that would require a new morphology and building-system modelling scope. The manuscript states this as a limitation and future-work direction.

Revisions made in the manuscript.

The relevant revisions appear in Sections 2.6, 3.5, 4.2, 4.3, Fig. 10, Table 6, and Supplementary Section S6.

Relevant revised manuscript and supporting evidence.

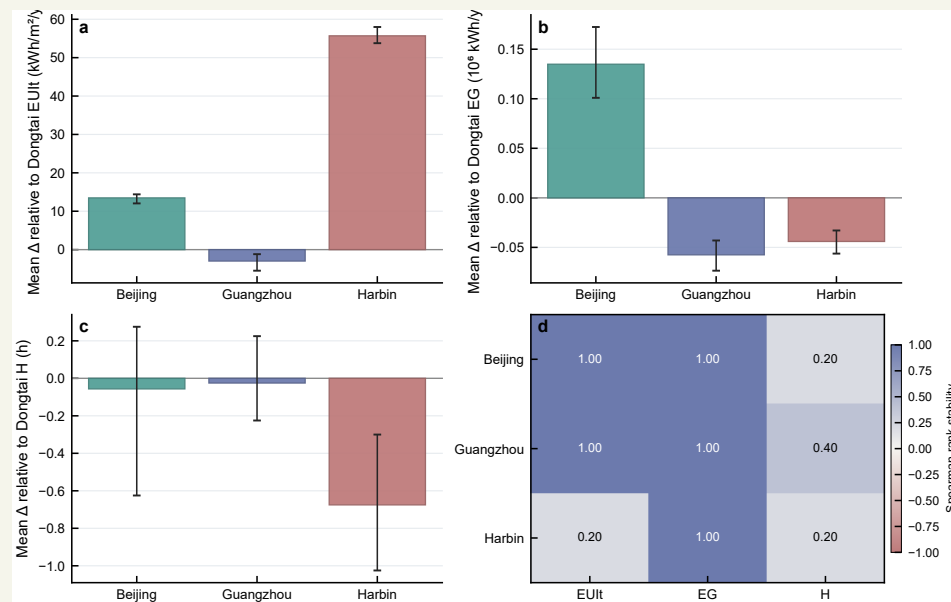
Main manuscript, Section 2.6 paragraph.

The baseline weather file used in the physical workflow serves as the reference condition for the cross-climate sensitivity analysis. The cross-climate sensitivity set then evaluates the same four block morphologies under three additional weather files representing Harbin, Beijing, and Guangzhou. The surrogate is not retrained, and the morphology set is not reselected after observing climate results.

Main manuscript, Section 3.5 paragraph.

The climate analysis in Fig. 10 evaluates four selected blocks under three additional weather files relative to the baseline weather. The demand response is strongly climate-dependent, especially under the severe-cold Harbin weather. EG ordering remains stable across the additional climates, suggesting that the rooftop-generation proxy is less sensitive to the weather substitution for this four-block set. H ordering is less stable, which is consistent with its dependence on solar geometry, obstruction, and weather-specific direct-sun conditions.

Main manuscript, Fig. 10.



Revised Fig. 10. Cross-climate sensitivity of four selected blocks relative to baseline weather.

Supplementary Material, Table S5 excerpt.

Station	ΔEUI_t	ΔEG	ΔH	Main interpretation
Beijing	13.467	0.135	-0.056	higher demand
Guangzhou	-2.972	-0.057	-0.025	lower demand
Harbin	55.681	-0.044	-0.675	strong demand increase

Revised Table S5 excerpt. ΔEUI_t is in $\text{kWh m}^{-2} \text{y}^{-1}$, ΔEG is in 10^6kWh y^{-1} , and ΔH is in h.

Supplementary Material, Section S6.3 paragraph.

The rank-stability results indicate target-specific climate sensitivity. EG has stable ordering across the three additional climates, with Spearman ρ equal to 1.0 in each case. H shows lower ordering stability, with Spearman ρ values of 0.2 for Beijing, 0.4 for Guangzhou, and 0.2 for Harbin. EUI_t ordering remains stable for Beijing and Guangzhou but changes under Harbin, where Spearman ρ is 0.2. These results support a limited climate-sensitivity interpretation for the selected four-block set.

Scope boundary.

The added analysis is a four-block climate sensitivity test. It is not a cross-climate surrogate-generalization study and does not add new building types.

■ Comment R1-15.

I think it would be great if the authors could write this paper for a broader audience.

Response.

✓ We rewrote the Introduction, added an Abbreviations block, reorganized the Methodology, and separated evaluation modes in Table 2 so that readers can follow the data source, optimizer setting, and interpretation boundary without relying on optimizer-specific assumptions.

Revisions made in the manuscript.

The relevant revisions appear in the Abstract, Abbreviations, Introduction, Table 2, Discussion, and Supplementary Information structure.

Relevant revised manuscript and supporting evidence.

Main manuscript, Introduction motivation excerpt.

Urban energy planning is increasingly shaped by the dual requirement of reducing building-sector emissions and maintaining reliable energy services. At the district and block scales, urban morphology controls floor area, envelope exposure, roof availability, obstruction, and solar access, which together affect both energy demand and distributed generation potential.

Main manuscript, contribution paragraph.

This study develops a surrogate-assisted comparison workflow for block-scale urban energy design. The contributions are threefold: the study establishes a morphology-to-performance workflow in which feasible residential block layouts are generated before 12 morphology descriptors are calculated; the study compares DDPG and NSGA-II under the same selected DNN surrogate and matched query budget; and the study connects surrogate-based optimization with repeated surrogate assessment, feasible-block projection, a 24-case physics-based cross-model stress test, and a four-block cross-climate sensitivity analysis.

Main manuscript, Table 2 excerpt.

Mode	Source	Output	Boundary
Analytic response generation	generated descriptors	EUI _t , EG, H	analytic surface
Surrogate prediction	12 descriptors	DNN outputs	selected checkpoint
Feasible projection	descriptor candidate	nearest block	nearest neighbour
Physical stress test	locked blocks	physical outputs	24 cases
Climate sensitivity	four blocks	weather deltas	no retraining

Revised Table 2 excerpt. Evaluation modes are separated by source, output, and interpretation boundary.

Scope boundary.

The clearer framing improves readability while preserving the same study scope.

► Reviewer 2.

✓ We thank Reviewer 2 for identifying the central issues around analytic approximation, physical-probe terminology, validation scope, sample generation, RL formulation, evaluation terminology, target formulas, and reproducibility. These comments shaped the largest part of the revision.

■ Comment R2-1.

The manuscript states that the dataset is generated through an "analytic approximation" of the original EnergyPlus/Radiance workflow rather than through direct simulation. Please provide reasons for this workflow.

Response.

✓ We clarified that the dataset is an analytic response-generation dataset used for controlled repeated optimizer comparison. This design gives a reproducible response surface for matching query budgets across DDPG and NSGA-II, while selected locked cases are later used for cross-model stress testing.

Revisions made in the manuscript.

The relevant revisions appear in Sections 2.1–2.3, Table 2, and Supplementary Section S1.4.

Relevant revised manuscript and supporting evidence.

Main manuscript, Section 2.2 paragraphs.

This module provides a controlled morphology-to-performance response surface for optimizer comparison. The nested 500/1000/1500/2000-sample pools are deterministic subsets of the same generated morphology pool under the recorded random seed and morphology-generation protocol. The physics-based case calculations are conducted subsequently on locked generated blocks to assess how the analytic target values relate to EnergyPlus- and Radiance-component outputs in selected cases.

The response directions follow the design objectives: lower EUI is preferred, while higher EG and H are preferred. The active 2000-sample pool is the largest nested sample regime used for the selected DNN surrogate. The full equations, station-bias convention, stochastic terms, and clipping ranges are given in Supplementary Section S1. Table 2 distinguishes analytic response assignment, surrogate prediction, feasible projection, and the post-optimization sensitivity analyses.

Main manuscript, Table 2 excerpt.

Mode	Source	Output	Boundary
Analytic response generation	generated descriptors	EUI _t , EG, H	analytic surface
Surrogate prediction	12 descriptors	DNN outputs	selected checkpoint
Feasible projection	descriptor candidate	nearest block	nearest neighbour
Physical stress test	locked blocks	physical outputs	24 cases
Climate sensitivity	four blocks	weather deltas	no retraining

Revised Table 2 excerpt. Evaluation modes are separated by source, output, and interpretation boundary.

Supplementary Material, Section S1.4 opening paragraph.

The analytic response generator is the target-assignment module that maps a generated morphology descriptor vector, denoted $\mathbf{x}$, to three target responses, denoted $\mathbf{y}^{ana} = [\text{EUI}_t, \text{EG}, H]^T$. The mapping uses explicit formulas that combine density, openness, aspect-ratio, sky-view, orientation, and open-space-position terms to represent demand, rooftop-generation, and sunlight-access responses.

Scope boundary.

The analytic dataset supports controlled repeated optimizer comparison. It is not direct annual EnergyPlus/Radiance simulation for all 2000 samples.

■ **Comment R2-2.**

The study repeatedly refers to "physical probe" and "physical-stack probe," but the description suggests that the probe is still simplified and only partially aligned with the original metrics.

Response.

✓ We replaced the stronger probe terminology with limited physics-based cross-model stress test. The manuscript now states that EUI_t, EG proxy, and H are completed for locked cases but that agreement is limited and the EG output is a GHI-based rooftop-PV proxy.

Revisions made in the manuscript.

The relevant revisions appear in Table 2, Sections 2.6 and 3.5, Table 6, Fig. 9, and Supplementary Sections S5.3–S5.4.

Relevant revised manuscript and supporting evidence.

Main manuscript, Table 2 excerpt.

Mode	Source	Output	Boundary
Analytic response generation	generated descriptors	EUI _t , EG, H	analytic surface
Surrogate prediction	12 descriptors	DNN outputs	selected checkpoint
Feasible projection	descriptor candidate	nearest block	nearest neighbour
Physical stress test	locked blocks	physical outputs	24 cases
Climate sensitivity	four blocks	weather deltas	no retraining

Revised Table 2 excerpt. Evaluation modes are separated by source, output, and interpretation boundary.

Main manuscript, Section 2.6 paragraph.

The limited physics-based cross-model stress test uses 24 locked cases. Eighteen direct feasible cases compare analytic response-generator values with physics-based outputs without optimizer projection, while six optimizer-linked cases decompose the path from descriptor-space candidate to nearest generated feasible block and then to physics-based result. The direct and optimizer-linked subsets answer different questions and are not merged into a single parity plot.

Main manuscript, Section 3.5 paragraph.

The physics-based cross-model stress test completed all locked cases and provides a case-level comparison between analytic target values and physics-based outputs. Figure 9 uses only the 18 direct feasible blocks, so the parity plots are not affected by optimizer projection. The three targets show limited metric and rank consistency in this case set.

Supplementary Material, Section S5.4 paragraphs.

The locked physical workflow did not produce annual roof-plane irradiance outputs for the direct stress-test cases. The unavailable roof-irradiance pathway is therefore kept separate from the completed EUI_t, EG proxy, and H calculations.

EG is summarized with the GHI-based rooftop-PV proxy in the stress-test table. This proxy preserves a generation-related comparison across the locked cases, but it should be interpreted as a simplified rooftop-PV indicator rather than a full annual roof-irradiance result.

Scope boundary.

The stress test is a limited cross-model comparison. It is not a complete physical ranking assessment.

■ **Comment R2-3.**

The validation evidence remains insufficient. Only a small-batch probe is used, and the physical results show noticeable deviations from surrogate predictions, especially for EUI_t and H.

Response.

✓ We agree and revised the interpretation accordingly. The manuscript now reports the deviations directly and uses them to bound the conclusions. It no longer treats the physical stress test as confirmation of optimizer ranking transfer.

Revisions made in the manuscript.

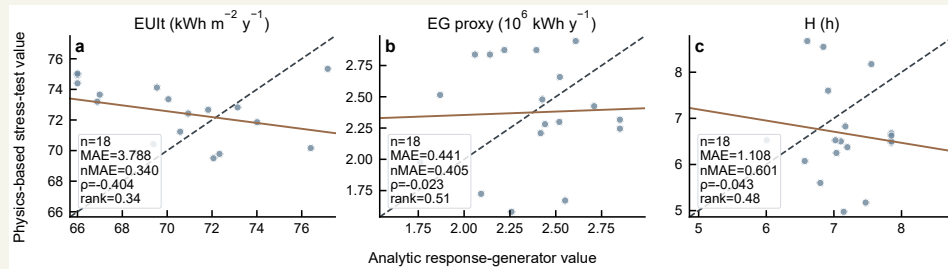
The relevant revisions appear in Sections 2.6, 3.5, 4.2, Fig. 9, Table 6, and Supplementary Section S5.

Relevant revised manuscript and supporting evidence.

Main manuscript, Section 3.5 paragraph.

The physics-based cross-model stress test completed all locked cases and provides a case-level comparison between analytic target values and physics-based outputs. Figure 9 uses only the 18 direct feasible blocks, so the parity plots are not affected by optimizer projection. The three targets show limited metric and rank consistency in this case set, with the weakest ordering stability appearing in the sunlight-related and demand-related comparisons. This result does not diminish the value of the stress test; instead, it clarifies the degree to which the analytic response surface and the physics-based calculations differ for selected generated blocks.

Main manuscript, Fig. 9.



Revised Fig. 9. Cross-model stress test for 18 direct generated feasible blocks.

Main manuscript and Supplementary Material, physical summary excerpt.

Target	Count	MAE	nMAE	Spearman ρ
EUIt	18	3.788	0.340	-0.404
EG GHI proxy	18	0.441	0.405	-0.023
H	18	1.108	0.601	-0.043

Revised Table 6 and Table S4 excerpt. The direct feasible stress-test cases show limited metric and rank consistency.

Supplementary Material, Section S5.2 paragraph.

The physics-based stress-test set contains 24 locked cases. The 18 direct feasible cases are used to compare analytic response-generator values with physics-based outputs without optimizer projection.

The six optimizer-linked cases decompose the path from descriptor-space candidate to nearest generated block and then to physics-based result. These two subsets should not be merged into a single parity plot because they answer different questions.

Scope boundary.

Metric agreement is weak and optimizer ranking transfer is unsupported.

■ **Comment R2-4.**

The manuscript claims that the final benchmark uses the most accurate 2000-sample surrogate selected from a 500/1000/1500/2000-sample scale study. However, the sample-generation strategy is not sufficiently described. The authors should specify whether sampling was random, Latin hypercube, grid-based, constraint-based, or derived from feasible morphology generation. The distribution and coverage of the 12-dimensional design space should also be shown.

Response.

✓ We added a full description of the feasible morphology generation sequence and descriptor coverage. The samples are generated from feasible block morphologies rather than from an independent 12-dimensional Latin hypercube or grid. We also added descriptor distributions, dependency checks, nearest-neighbour summaries, duplicate checks, and PCA coverage.

Revisions made in the manuscript.

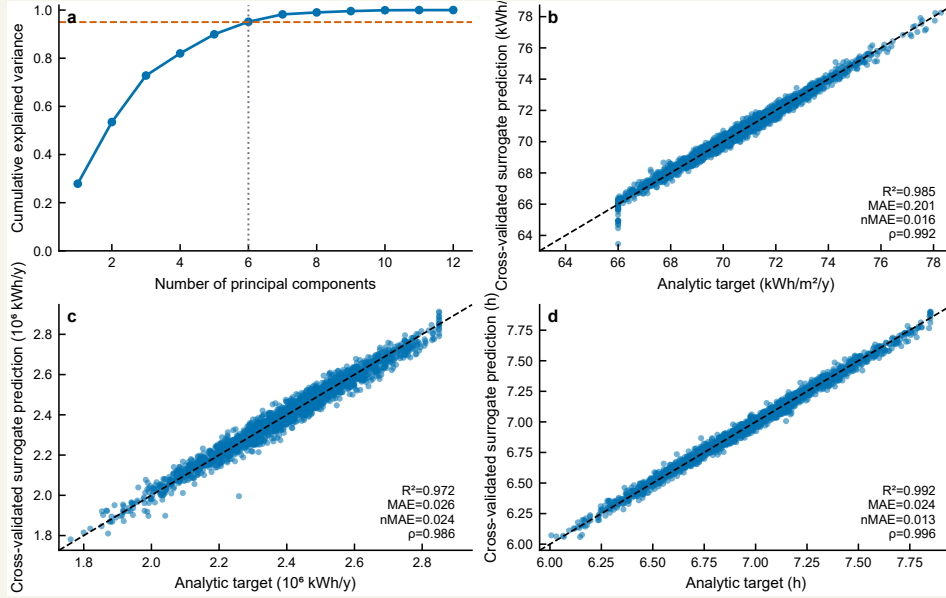
The relevant revisions appear in Section 2.1, Section 3.1, Table 1, Fig. 4, and Supplementary Sections S1.1–S1.5.

Relevant revised manuscript and supporting evidence.

Main manuscript, Section 2.1 paragraphs.

The morphology generator first constructs feasible block layouts and only then calculates descriptor values. Each block is represented as a 240 m × 240 m square divided into a 3 × 3 grid of 80 m × 80 m land units. One cell is assigned as open space, and the remaining eight cells are populated with residential prototypes. For each generated block, the generator samples the open-space position, building prototype, number of floors, and orientation angle. The orientation angle is sampled within -45 deg to 45 deg.

Main manuscript, Fig. 4.



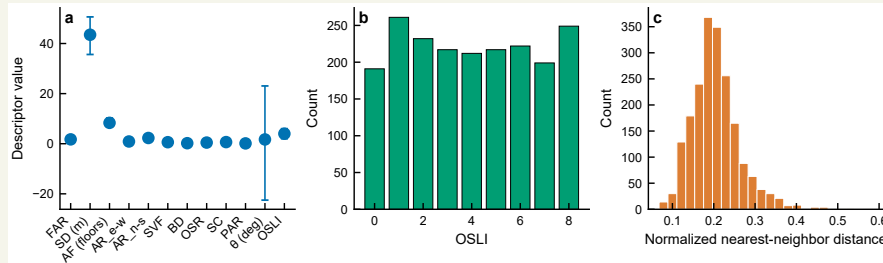
Revised Fig. 4. Descriptor coverage and cross-validated surrogate predictions.

Supplementary Material, Section S1.2 paragraph.

For each generated block, the open-space index is sampled from 0 to 8, theta is sampled uniformly from -45 deg to 45 deg, and each populated cell receives a randomly sampled prototype and a floor count sampled within that prototype range. Descriptor values are calculated only after this geometry-generation step. This order keeps the generated feasible block as the primary object and treats the 12 surrogate inputs as derived morphology descriptors.

Supplementary Material, Table S2 and Fig. S1 excerpts.

Check	Value	Interpretation
Exact duplicate count	0	no exact duplicate descriptor rows
Nearest-neighbour distance median	0.199	sparse normalized descriptor space
PCA components for 95% variance	6	structured descriptor space



Scope boundary.

The generated pool is a feasible-morphology descriptor pool, not an unconstrained independent 12-dimensional grid.

■ **Comment R2-5.**

The RL formulation remains questionable because the problem is essentially a static black-box optimization problem rather than a naturally sequential decision-making process. The state transition is determined by the surrogate response to a morphology vector, and the sequential meaning of episodes and steps is unclear.

Response.

✓ We agree and rewrote the formulation accordingly. The revised manuscript states that DDPG is used as serialized static black-box search on a surrogate response surface. The state is the normalized target output, the action is an absolute 12-descriptor vector, and the episode is a 40-query training horizon.

Revisions made in the manuscript.

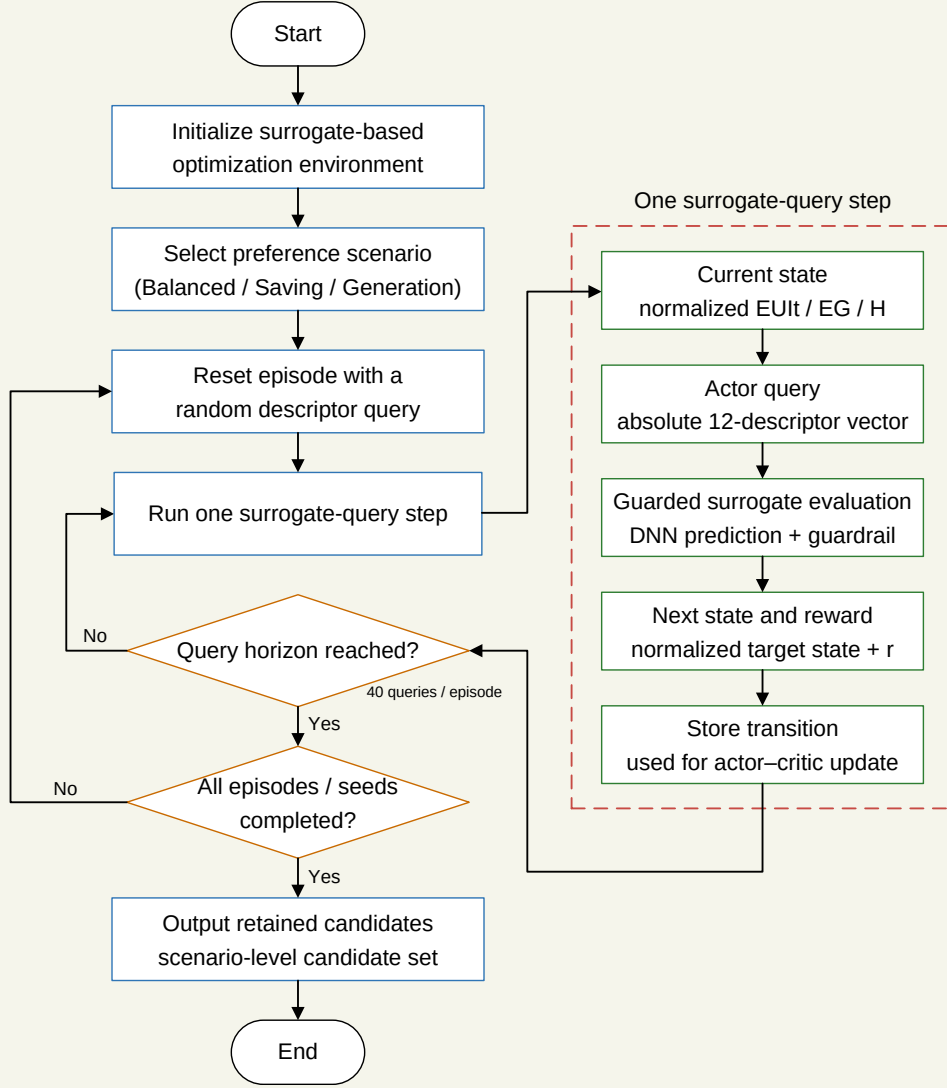
The relevant revisions appear in Sections 1.2, 2.4, and 3.2, Fig. 2, Fig. 3, Fig. 6, and Supplementary Section S3.1.

Relevant revised manuscript and supporting evidence.

Main manuscript, Section 2.4 paragraph.

The DDPG environment serializes repeated queries to a static guarded surrogate. At the beginning of each episode, a random descriptor query is evaluated to construct the initial normalized three-target state. At each subsequent step, the actor outputs an absolute 12-dimensional normalized descriptor action. This action is not an incremental modification of the preceding block. The action is denormalized to descriptor bounds, evaluated by the guarded surrogate, and converted to the next normalized target state and scalar reward. The 40-step horizon therefore functions as a fixed sequence of policy-training queries on a static surrogate response surface.

Main manuscript, Fig. 2.



Revised Fig. 2. Serialized DDPG surrogate-query workflow. Each episode contains 40 absolute descriptor queries.

Main manuscript, reward equations.

$$z_i = \frac{y_i - y_i^{\min}}{y_i^{\max} - y_i^{\min}}.$$

$$\mathbf{u} = [0, 1, 1]^T.$$

$$d_w = \frac{\|\mathbf{w} \odot (\mathbf{z} - \mathbf{u})\|_2}{\|\mathbf{w}\|_2}, \quad R = 1 - d_w.$$

Supplementary Material, Section S3.1 paragraph.

Each DDPG scenario uses 20 seeds, 600 episodes, and 40 surrogate interactions per episode. The seed-level summaries record best reward, final reward, plateau episode, and best-to-final regression. These quantities describe policy-training behaviour under serialized surrogate queries on a static surrogate response surface.

Scope boundary.

The sequence represents repeated black-box surrogate queries, not physical-time morphology evolution.

■ Comment R2-6.

The manuscript uses several terms that need clearer distinction, including "analytic response-generation workflow," "independent analytic reevaluation," "physical probe," "full physical stack," and "publication-grade physical closure." These terms are currently confusing. A table comparing these evaluation modes would improve transparency.

Response.

✓ We removed confusing stronger terminology and added a compact evaluation-mode table. The revised terminology is analytic response generation, surrogate prediction, feasible projection, physics-based cross-model stress test, and cross-climate sensitivity.

Revisions made in the manuscript.

The relevant revisions appear in Table 2 and throughout Sections 2.2, 2.6, 3.5, and 4.2.

Relevant revised manuscript and supporting evidence.

Main manuscript, Table 2 excerpt.

Mode	Source	Output	Boundary
Analytic response generation	generated descriptors	EUI, EG, H	analytic surface
Surrogate prediction	12 descriptors	DNN outputs	selected checkpoint
Feasible projection	descriptor candidate	nearest block	nearest neighbour
Physical stress test	locked blocks	physical outputs	24 cases
Climate sensitivity	four blocks	weather deltas	no retraining

Revised Table 2 excerpt. Evaluation modes are separated by source, output, and interpretation boundary.

Main manuscript, Section 2.6 paragraph.

Descriptor-space optimizer outputs are treated first as candidate descriptors. Each retained descriptor-space candidate is therefore mapped to the nearest generated feasible block in normalized descriptor space. This mapping quantifies projection distance, duplicate collapse, and changes in HV and IGD under the fixed normalized reference front. The projection step is a representation check that precedes morphology-level interpretation.

Main manuscript, Discussion boundary paragraph.

These two sensitivity analyses do not convert the descriptor-space comparison into a physical ranking

result. They instead define where the analytic response surface, generated feasible blocks, and selected physical workflows agree only partially, which is the boundary needed for early-stage design-support interpretation.

Scope boundary.

Analytic response generation, surrogate prediction, feasible projection, physical stress testing, and climate sensitivity are distinct evaluation modes.

■ **Comment R2-7.**

The objective definitions require further clarification. EUI_t, EG, and H are said to conceptually correspond to EnergyPlus, rooftop PV, and sunlight-hour calculations, but the actual values are generated analytically. The authors should provide clear formulas or workflows for calculating each target in the generated dataset.

Response.

✓ We added the analytic response equations in Supplementary Section S1.4 and summarized their role in the main manuscript. The revised text states that lower EUI_t is preferred and higher EG and H are preferred.

Revisions made in the manuscript.

The relevant revisions appear in Section 2.2, Table 2, and Supplementary Section S1.4.

Relevant revised manuscript and supporting evidence.

Main manuscript, Section 2.2 paragraph.

The response directions follow the design objectives: lower EUI_t is preferred, while higher EG and H are preferred. The active 2000-sample pool is the largest nested sample regime used for the selected DNN surrogate. The full equations, station-bias convention, stochastic terms, and clipping ranges are given in Supplementary Section S1. Table 2 distinguishes analytic response assignment, surrogate prediction, feasible projection, and the post-optimization sensitivity analyses.

Supplementary Material, Section S1.4 equations.

$$p_{\theta} = |\theta|/45, \quad a_m = 0.5(AR_{ew} + AR_{ns}),$$

$$s = \exp\left(-\frac{(FAR - 1.7)^2}{0.6}\right) (0.45 + SVF)(0.3 + OSR), \quad p_d = \max(FAR - 2.8, 0).$$

$$\begin{aligned}
EUIt^* &= 78.0 + 10.0BD + 4.3PAR + 2.8a_m + 0.022SD + 0.8SC + 2.1p_\theta \\
&\quad - 6.4OSR - 5.8SVF - 0.9AF - 4.5s + \epsilon_{EUIt}, \\
EG^* &= 0.96 + 0.28FAR + 0.64SVF + 0.52OSR - 0.19BD - 0.12a_m - 0.18p_\theta \\
&\quad + 0.26OSLI^* + 0.78s - 0.1p_d + \epsilon_{EG}, \\
H^* &= 6.45 + 1.1SVF + 0.55OSR + 0.48OSLI^* + 0.24s - 0.7p_\theta - 0.45PAR - 0.24a_m + \epsilon_H. \\
EUIt &= \text{clip}(EUIt^*, 66.0, 96.0), \quad EG = \text{clip}(EG^*, 1.2, 2.85), \quad H = \text{clip}(H^*, 6.0, 7.85).
\end{aligned}$$

Scope boundary.

These formulas define analytic target assignment for the generated dataset. They are not direct EnergyPlus/Radiance outputs.

■ **Comment R2-8.**

The paper should provide code, data, or at least detailed reproducibility information.

Response.

✓ We revised the Data and code availability statement to identify the public repository and the planned processed datasets, result tables, and reproducibility notes. We also keep credentials and private raw artifacts out of the public release scope.

Revisions made in the manuscript.

The relevant revision appears in the Data and code availability section.

Relevant revised manuscript and supporting evidence.

Main manuscript, Data and code availability.

The code, processed datasets, and result tables supporting the numerical analyses will be made available through the project repository at <https://github.com/haolinwang1313/DRL-Driven-Optimization>. The public release will include the scripts required to regenerate the analytic response dataset, surrogate assessment summaries, optimizer comparison tables, feasible-projection results, physics-based stress-test summaries, and cross-climate sensitivity results.

Scope boundary.

The public release package is limited to sanitized code, processed data, result tables, and reproducibility notes; it does not include credentials or raw private artifacts.

► Reviewer 3.

✓ We thank Reviewer 3 for the comments on workflow-figure spelling, utility interpretation, reward formulation, unit consistency, and analytic/physical wording. These comments helped align the figures, equations, and claims with the revised evidence.

■ Comment R3-1.

Correct spelling and wording errors in the workflow figures. For example, Fig. 1 still contains visible typos such as "loactions," "Acotr network," and "Critic network," which should be revised to "locations," "Actor network," and "Critic network."

Response.

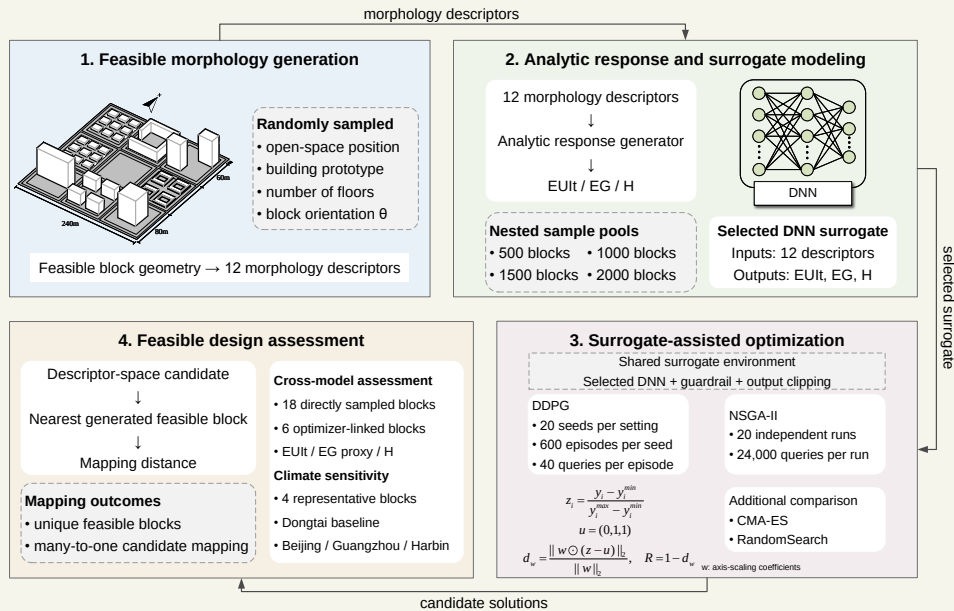
✓ We corrected and replaced the workflow and DDPG figures. The revised figures use corrected labels, including locations, Actor network, and Critic network.

Revisions made in the manuscript.

The relevant revisions appear in Fig. 1, Fig. 3, and their captions.

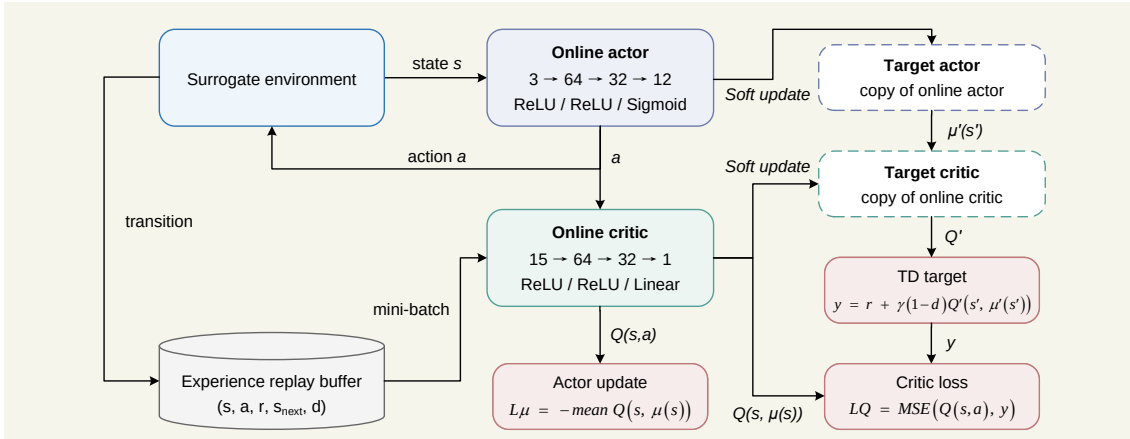
Relevant revised manuscript and supporting evidence.

Main manuscript, Fig. 1.



Revised Fig. 1. Workflow linking feasible morphology generation, analytic response assignment, surrogate-assisted optimization, feasible-block projection, and sensitivity analysis.

Main manuscript, Fig. 3.



Revised Fig. 3. DDPG actor-critic architecture and target-network update structure used in the surrogate-query environment.

Scope boundary.

These are presentation and terminology corrections; they do not change the scientific results.

■ Comment R3-2.

Verify the consistency between Fig. 9(d) and the stated benchmark conclusion. The text states that NSGA-II remains stronger than all DDPG scenarios, but the utility panel should be carefully checked to ensure that the plotted curves, legend, and utility direction support this conclusion.

Response.

✓ We reworked the optimizer comparison so that it no longer relies on an ambiguous universal superiority statement. The revised benchmark separates fixed-domain post-hoc utility, equal-size descriptor-space HV and IGD, output contracts, full-archive interpretation, and feasible-block projection.

Revisions made in the manuscript.

The relevant revisions appear in Sections 3.3 and 3.4, Table 5, Fig. 7, Fig. 8, and Supplementary Section S3.

Relevant revised manuscript and supporting evidence.

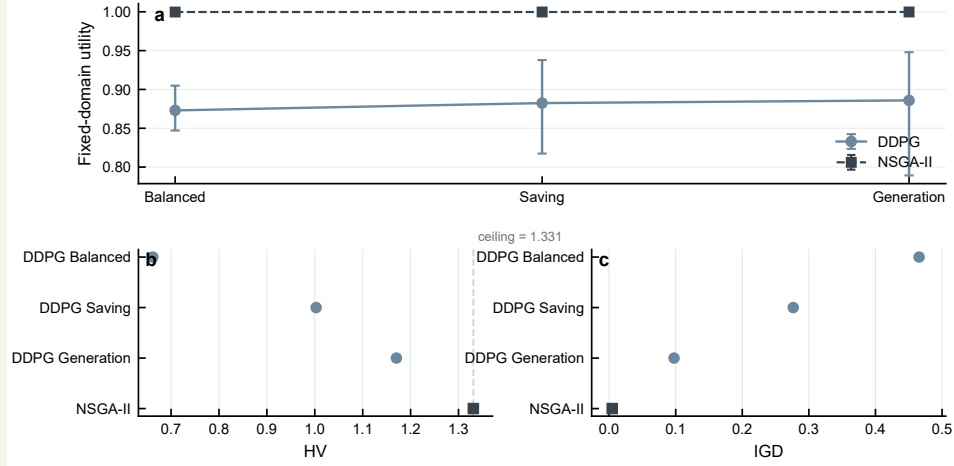
Main manuscript, Section 3.3 paragraphs.

Table 5 summarizes descriptor-space performance under the fixed normalized reference front, while Fig. 7 separates fixed-domain utility from equal-size archive metrics. NSGA-II is close to the HV ceiling under the fixed reference and remains the strongest method in the equal-size HV and IGD comparison. Among the three DDPG settings, the Generation-focused scenario gives the strongest descriptor-space archive metrics, followed by the Energy Saving and Balanced settings.

The comparison should be interpreted through the retained-output structure. DDPG contributes one best scalarized candidate from each seed, whereas NSGA-II contributes final populations from repeated

runs. The equal-size summary reduces, but does not eliminate, this structural difference. Therefore, the descriptor-space result supports a method-comparison statement under the selected surrogate and fixed reference; it does not by itself describe feasible-block diversity.

Main manuscript, Fig. 7.



Revised Fig. 7. Descriptor-space DDPG and NSGA-II comparison under fixed-domain utility and equal-size retained outputs.

Main manuscript, Table 5 excerpt.

Method	Scenario	HV	IGD	Note
DDPG	Balanced	0.661912	0.465522	scalarized output
DDPG	Saving	1.002757	0.276529	scalarized output
DDPG	Generation	1.170428	0.097551	scalarized output
NSGA-II	equal-size 20	1.330995	0.004710	common retained size
NSGA-II	full archive	1.330999	0.004947	2000 rows

Revised Table 5 excerpt. Descriptor-space metrics use a fixed normalized reference front.

Main manuscript, Section 3.4 paragraph.

The projection analysis shows that descriptor-space coverage and feasible-morphology diversity are distinct. NSGA-II retains 2000 descriptor candidates, yet nearest-neighbour projection maps them to 51 unique generated feasible blocks.

Scope boundary.

NSGA-II is stronger in descriptor-space equal-size HV and IGD under the fixed reference; this is not a universal optimizer-superiority claim.

■ Comment R3-3.

Recheck the mathematical form of the weighted-distance reward in Eq. (5). The current expression appears to apply weights directly to the normalized objective values; if the intended formulation is a standard weighted Euclidean distance, the equation should be revised accordingly.

Response.

✓ We corrected the reward expression. The revised manuscript now defines normalized target coordinates, the ideal vector, the weighted normalized distance, and the reward. It also states that the previous large-offset reward expression has been removed and that training reward is distinct from post-hoc utility.

Revisions made in the manuscript.

The relevant revisions appear in Section 2.4 and Fig. 3.

Relevant revised manuscript and supporting evidence.

Main manuscript, Section 2.4 reward equations.

$$z_i = \frac{y_i - y_i^{\min}}{y_i^{\max} - y_i^{\min}},$$

$$\mathbf{u} = [0, 1, 1]^T.$$

$$d_w = \frac{\|\mathbf{w} \odot (\mathbf{z} - \mathbf{u})\|_2}{\|\mathbf{w}\|_2}, \quad R = 1 - d_w.$$

Main manuscript, preference-weight sentence.

The preference weights follow the target order [EUIt, EG, H]: Balanced uses $\mathbf{w} = [1/3, 1/3, 1/3]^T$, Energy Saving uses $\mathbf{w} = [0.6, 0.2, 0.2]^T$, and Energy Generation uses $\mathbf{w} = [0.2, 0.6, 0.2]^T$.

Main manuscript, reward boundary sentence.

This reward is separate from the fixed-domain post-hoc utility used later for benchmark comparison. The fixed-domain utility is used for post-hoc comparison, while R is the scalar signal used during serialized DDPG training.

Scope boundary.

The earlier large-offset reward expression is removed. The reward is distinct from the fixed-domain post-hoc utility.

■ Comment R3-4.

Further standardize units in figure axes and table headings. Terms such as EUIt, EG, H, residuals, and morphology factors should consistently include units where applicable, for example "EUIt (kWh/m²/y)," "EG (10⁶ kWh/y)," and "H (h)."

Response.

✓ We standardized units in figure axes, table headings, table notes, and descriptor definitions. The manuscript consistently reports EUIt in kWh m⁻² y⁻¹, EG in 10⁶ kWh y⁻¹, H in hours, theta in degrees, and SD in metres where applicable.

Revisions made in the manuscript.

The relevant revisions appear in Table 1, Table 3, Table 5, Table 6, Figs. 4–10, and Supplementary tables and captions.

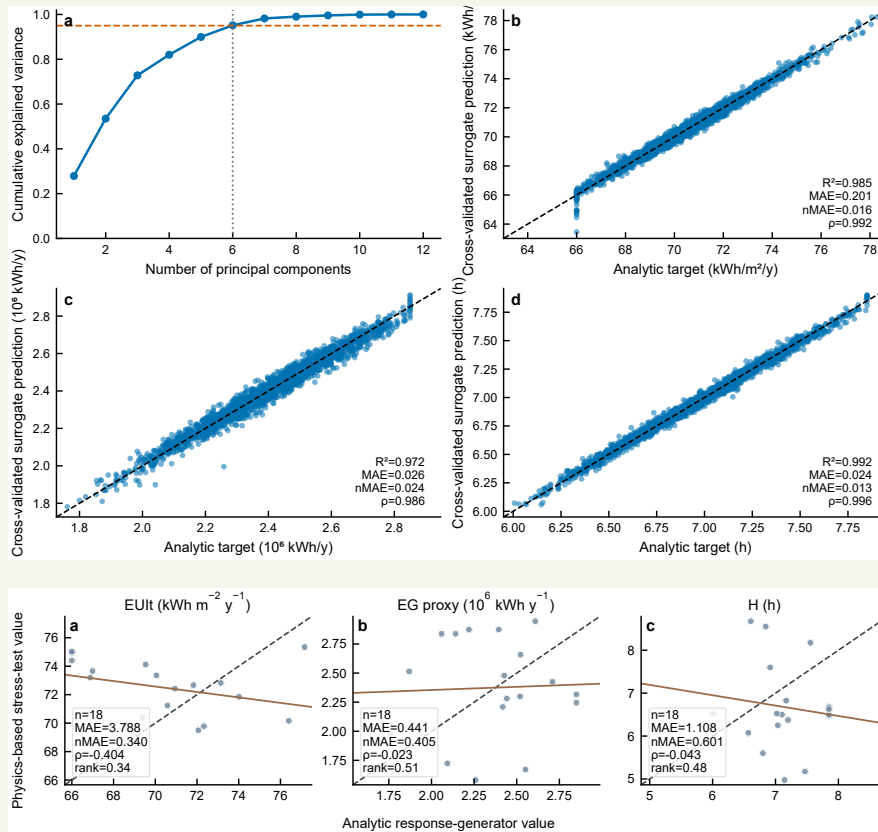
Relevant revised manuscript and supporting evidence.

Main manuscript, Table 1 descriptor units excerpt.

Descriptor	Unit or type	Construction	Interpretation
FAR	–	$A_{\text{gross}}/A_{\text{block}}$	density
SD	m	$h_{\text{max}} - \bar{h}$	height variation
AF	floors	$A_{\text{gross}}/A_{\text{foot}}$	linked to FAR, BD
AR_{ew}	–	$\bar{h}/W_{ew}$	east-west canyon
AR_{ns}	–	$\bar{h}/W_{ns}$	north-south canyon
SVF	–	analytic openness proxy	sky-view openness
BD	–	$A_{\text{foot}}/A_{\text{block}}$	footprint density
OSR	–	$A_{\text{open}}/A_{\text{gross}}$	open-space ratio
SC	–	$A_{\text{env}}/A_{\text{gross}}$	envelope exposure
PAR	m ⁻¹	$P_{\text{sum}}/A_{\text{foot}}$	compactness
θ	deg	sampled in $[-45, 45]$	orientation
OSLI	integer	0–8	open-space cell

Revised Table 1 excerpt. The 12 quantities are derived morphology descriptors, not independent design variables.

Main manuscript, Fig. 4 and Fig. 9 unit-bearing evidence.



Revised figures. Target units are attached to axes and table headings where applicable.

Supplementary Material, Table S5 note.

ΔEUIIt is in $\text{kWh m}^{-2} \text{y}^{-1}$, ΔEG is in 10^6kWh y^{-1} , and ΔH is in h. Positive values denote increases relative to the baseline weather file.

Scope boundary.

The unit edits improve traceability and do not modify canonical values.

Comment R3-5.

Make the wording about simulation, analytic approximation, and physical probing fully consistent. Since the manuscript states that the dataset is based on an analytic approximation rather than direct full-stack EnergyPlus/Radiance simulations, all related phrases should avoid implying that the final benchmark was performed using direct physical simulation.

Response.

✓ We revised the terminology throughout the manuscript. The 2000-sample benchmark is now described as an analytic response-generation dataset, while selected case calculations are described as a limited physics-

based cross-model stress test.

Revisions made in the manuscript.

The relevant revisions appear in the Abstract, Sections 1–4, Table 2, Table 6, and Supplementary Sections S1 and S5.

Relevant revised manuscript and supporting evidence.

Main manuscript, Abstract excerpt.

Feasible block layouts are generated first and converted into 12 morphology descriptors. A morphology-to-performance response surface then assigns three targets to each sample: Energy Use Intensity, Energy Generation, and winter sunlight hours.

Main manuscript, Table 2 excerpt.

Mode	Source	Output	Boundary
Analytic response generation	generated descriptors	EUI _t , EG, H	analytic surface
Surrogate prediction	12 descriptors	DNN outputs	selected checkpoint
Feasible projection	descriptor candidate	nearest block	nearest neighbour
Physical stress test	locked blocks	physical outputs	24 cases
Climate sensitivity	four blocks	weather deltas	no retraining

Revised Table 2 excerpt. Evaluation modes are separated by source, output, and interpretation boundary.

Main manuscript, Discussion boundary paragraph.

These two sensitivity analyses do not convert the descriptor-space comparison into a physical ranking result. They instead define where the analytic response surface, generated feasible blocks, and selected physical workflows agree only partially, which is the boundary needed for early-stage design-support interpretation.

Supplementary Material, Section S5.4 paragraph.

EG is summarized with the GHI-based rooftop-PV proxy in the stress-test table. This proxy preserves a generation-related comparison across the locked cases, but it should be interpreted as a simplified rooftop-PV indicator rather than a full annual roof-irradiance result.

Scope boundary.

The 2000-sample benchmark uses analytic response generation; direct physical calculations are limited to locked stress-test cases.

► **Reviewer 4.**

✓ We thank Reviewer 4 for comments on study necessity, climate-zone effects, energy-system expansion, best-scenario definition, and comparison with existing urban energy planning approaches. The revised manuscript strengthens the motivation while preserving the current model scope.

■ **Comment R4-1.**

There are plenty of studies related to block-scale urban energy planning, the necessity of this study needs to be further highlighted.

Response.

✓ We revised the Introduction to clarify that the study is not simply another optimizer application. The revised necessity is methodological: optimizer ranking in early-stage block-scale design depends on surrogate conditioning, representation, retained outputs, and feasible projection.

Revisions made in the manuscript.

The relevant revisions appear in Sections 1.1–1.3 and the Discussion.

Relevant revised manuscript and supporting evidence.

Main manuscript, Motivation paragraphs.

Block morphology influences energy performance through coupled physical mechanisms. Density and height alter heating and cooling demand by changing envelope exposure, mutual shading, and micro-climatic context. Solar access and daylight availability depend on roof exposure, sky-view conditions, block orientation, and open-space arrangement. Recent large-scale studies further show that urban spatial structure can shape building energy demand across broad urban areas and under extreme climate events, which makes morphology-sensitive design relevant beyond isolated building cases.

The computational challenge is that morphology-sensitive energy analysis becomes expensive when thousands of candidate blocks, multiple optimizer seeds, and several preference settings must be evaluated. Simulation-based urban form generation and building-performance optimization provide a strong basis for this task, but repeated full physical simulation remains costly in search loops.

Main manuscript, gap paragraph.

The remaining gap is methodological and representation-oriented. Urban energy optimization has established the value of morphology-aware design, surrogate modelling has reduced the cost of repeated evaluation, and reinforcement learning has proved useful in several energy-operation settings. What remains less clear is how policy learning and Pareto search should be compared when both operate on a shared surrogate response surface, when the surrogate inputs are descriptors derived from generated

blocks, and when continuous outputs require projection before they can be interpreted as feasible morphologies.

Scope boundary.

The contribution is a representation-aware design-support workflow, not a claim that one optimizer is universally superior across urban energy planning problems.

Local references.

- Perera, A. T. D., Javanroodi, K., et al. (2023). Challenges resulting from urban density and climate change for the EU energy transition. *Nature Energy*, 8(4), 397–412. <https://doi.org/10.1038/s41560-023-01232-9>
- Shi, Z., Fonseca, J. A., & Schlueter, A. (2017). A review of simulation-based urban form generation and optimization for energy-driven urban design. *Building and Environment*, 121, 119–129. <https://doi.org/10.1016/j.buildenv.2017.05.006>
- Westermann, P., & Evins, R. (2019). Surrogate modelling for sustainable building design: A review. *Energy and Buildings*, 198, 170–186. <https://doi.org/10.1016/j.enbuild.2019.05.057>

■ **Comment R4-2.**

The selected location for modelling is in a specific climate zone. How this specific climate condition may affect the modelling and how the findings could be translated into other climate zones? This is not explained clearly.

Response.

✓ We added a limited cross-climate sensitivity analysis using Harbin, Beijing, and Guangzhou weather files for the same four selected block morphologies. The revision presents this as sensitivity analysis, not as general transferability.

Revisions made in the manuscript.

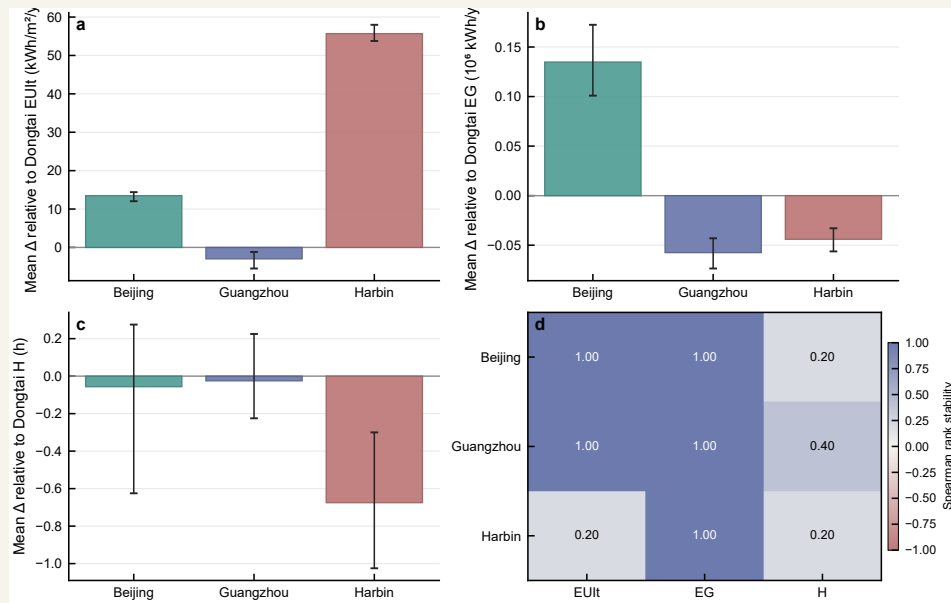
The relevant revisions appear in Sections 2.6, 3.5, and 4.2, Fig. 10, Table 6, and Supplementary Section S6.

Relevant revised manuscript and supporting evidence.

Main manuscript, Section 2.6 paragraph.

The baseline weather file used in the physical workflow serves as the reference condition for the cross-climate sensitivity analysis. The cross-climate sensitivity set then evaluates the same four block morphologies under three additional weather files representing Harbin, Beijing, and Guangzhou. The surrogate is not retrained, and the morphology set is not reselected after observing climate results.

Main manuscript, Fig. 10.



Revised Fig. 10. Cross-climate sensitivity of four selected blocks relative to baseline weather.

Supplementary Material, Section S6.1 paragraph.

The baseline weather file corresponds to the station used for the case-study physical workflow. The cross-climate sensitivity analysis evaluates the same four generated feasible blocks under Harbin, Beijing, and Guangzhou weather. The climate buckets are severe cold for Harbin, cold for Beijing, and hot-summer warm-winter for Guangzhou. The TMY period is 2009–2023 for these weather files.

Supplementary Material, Table S5 excerpt.

Station	ΔEUIt	ΔEG	ΔH	Main interpretation
Beijing	13.467	0.135	-0.056	higher demand
Guangzhou	-2.972	-0.057	-0.025	lower demand
Harbin	55.681	-0.044	-0.675	strong demand increase

Revised Table S5 excerpt. ΔEUIt is in $\text{kWh m}^{-2} \text{ y}^{-1}$, ΔEG is in 10^6 kWh y^{-1} , and ΔH is in h.

Supplementary Material, Section S6.3 paragraph.

EG has stable ordering across the three additional climates, with Spearman ρ equal to 1.0 in each case. H shows lower ordering stability, with Spearman ρ values of 0.2 for Beijing, 0.4 for Guangzhou, and 0.2 for Harbin. EUIt ordering remains stable for Beijing and Guangzhou but changes under Harbin, where Spearman ρ is 0.2.

Scope boundary.

This is a limited four-block cross-climate physical sensitivity analysis, not cross-climate surrogate

generalization.

■ Comment R4-3.

How the different sources of energy potentially affect the urban energy planning? This is specifically the case while China has rapidly developed the EV market. Can the distributed energy systems affect the urban energy planning?

Response.

✓ We clarified the current objective scope and added storage, EV charging, and distributed-energy interactions as future-work directions. The current study considers demand, rooftop-generation proxy, and winter sunlight hours; it does not model district-level grid exchange or EV charging.

Revisions made in the manuscript.

The relevant revision appears in Section 4.3.

Relevant revised manuscript and supporting evidence.

Main manuscript, Section 4.3 paragraph.

Several limitations remain. The optimizer comparison is conducted on an analytic response-generation dataset, while direct physical calculations are available for the locked stress-test cases. The 12 surrogate inputs are morphology-derived descriptors rather than independent unconstrained design variables. The DDPG horizon is best understood as a policy-training query sequence on a static surrogate. The cross-model stress test indicates limited rank consistency in the tested case set, and the climate analysis covers four blocks and three additional climates. Future work should connect optimization to a feasible morphology decoder, expand the physical simulation set, incorporate storage, EV charging, and distributed-energy interactions, and train climate-aware surrogates across multiple regions.

Main manuscript, objective scope excerpt.

The workflow generates feasible residential block morphologies, converts them into 12 morphology descriptors, assigns EUI, EG, and H through an explicit morphology-to-performance response surface, and trains a selected DNN surrogate from the 2000-sample regime of a nested scale study.

Scope boundary.

Storage, EV charging, district-level grid exchange, and distributed-energy operation are not modelled in the current objective system.

■ Comment R4-4.

It is not very clear what is considered as "best scenario" for optimization process.

Response.

✓ We removed ambiguous best-scenario wording and defined the comparison criteria explicitly. The revised manuscript separates best scalarized DDPG candidate, retained-output contract, fixed-domain post-hoc utility, descriptor-space HV and IGD, and projected feasible blocks.

Revisions made in the manuscript.

The relevant revisions appear in Sections 2.5, 3.3, and 3.4, Tables 4–5, and Figs. 7–8.

Relevant revised manuscript and supporting evidence.

Main manuscript, Table 4 excerpt.

Method	Runs	Queries/run	Retained output
DDPG	20 seeds	24,000	best scalarized point
NSGA-II	20 runs	24,000	final populations
CMA-ES	20 runs	24,000	supplementary archive
RandomSearch	20 runs	24,000	supplementary archive

Main manuscript, Table 5 excerpt.

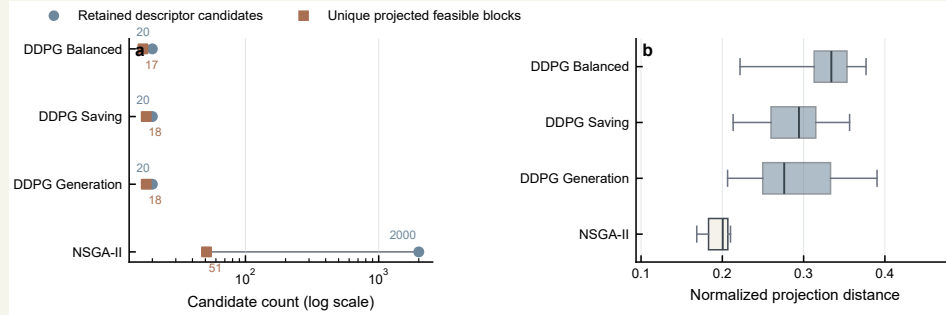
Method	Scenario	HV	IGD	Note
DDPG	Balanced	0.661912	0.465522	scalarized output
DDPG	Saving	1.002757	0.276529	scalarized output
DDPG	Generation	1.170428	0.097551	scalarized output
NSGA-II	equal-size 20	1.330995	0.004710	common retained size
NSGA-II	full archive	1.330999	0.004947	2000 rows

Revised Table 5 excerpt. Descriptor-space metrics use a fixed normalized reference front.

Main manuscript, Section 3.4 paragraph.

Projection distance adds a second layer of interpretation. The DDPG scenarios retain fewer descriptor candidates than NSGA-II, but their projected candidates are generally farther from the generated feasible blocks. In contrast, NSGA-II shows stronger descriptor-space coverage and shorter projection distances, yet its candidates still collapse to a much smaller feasible-block set.

Main manuscript, Fig. 8.



Revised Fig. 8. Descriptor-space candidate compression and projection distance after nearest-feasible-block mapping.

Scope boundary.

Best depends on criterion, retained-output contract, and projection stage.

■ Comment R4-5.

More discussions are required to compare the proposed urban energy planning framework with existing approaches to validate its advantages.

Response.

✓ We expanded the literature review and Discussion to compare the workflow with simulation-based optimization, surrogate-assisted optimization, NSGA-II Pareto archives, DRL operational-control studies, and neighbouring derivative-free methods. The revised discussion states the advantage narrowly: transparent separation of surrogate search, feasible projection, and selected sensitivity analysis.

Revisions made in the manuscript.

The relevant revisions appear in Sections 1.2, 4.1, and 4.3.

Relevant revised manuscript and supporting evidence.

Main manuscript, literature review paragraph.

Parametric design studies show that geometry generation can support energy- and daylight-oriented design exploration. Research on parametric daylighting and low-energy design demonstrates that early-stage geometry and envelope choices can be formalized as searchable design spaces. Evolutionary multi-objective optimization remains widely used because it provides explicit trade-off archives for conflicting objectives. NSGA-II and related methods have been applied to envelope design, photovoltaic-oriented inverse urban design, urban block optimization, and energy-efficient shading design.

Main manuscript, DRL comparison paragraph.

In the energy domain, DRL has been widely adopted for HVAC control, building operation,

distribution-network control, and renewable-building operation. These applications are primarily operational control problems in which states evolve through time. Static urban morphology optimization differs from that setting because each query is an absolute candidate description evaluated by a surrogate response surface.

Main manuscript, Discussion paragraph.

For early-stage urban energy design, the workflow is useful because it separates fast descriptor-space screening from feasible-block projection and sensitivity analysis. A surrogate optimizer can identify promising descriptor regions, while projection and selected physics-based calculations help determine how those regions relate to generated block geometries. This separation makes the workflow more transparent for design-support use.

Scope boundary.

The advantage is transparent representation-aware interpretation, not comprehensive superiority over existing approaches.

Local references.

- Nguyen, A.-T., Reiter, S., & Rigo, P. (2014). A review on simulation-based optimization methods applied to building performance analysis. *Applied Energy*, 113, 1043–1058. <https://doi.org/10.1016/j.apenergy.2013.08.061>
- Zhao, Z., Li, H., & Wang, S. (2025). Surrogate-assisted coordinated design optimization of building and microclimate considering their mutual impacts. *Applied Energy*, 383. <https://doi.org/10.1016/j.apenergy.2025.125374>
- Ascione, F., De Masi, R. F., et al. (2016). Optimization of building envelope design for nZEBs in Mediterranean climate: Performance analysis of a residential case study. *Applied Energy*, 183, 938–957. <https://doi.org/10.1016/j.apenergy.2016.09.027>
- Du, Y., Zandi, H., et al. (2021). Intelligent multi-zone residential HVAC control strategy based on deep reinforcement learning. *Applied Energy*, 281. <https://doi.org/10.1016/j.apenergy.2020.116117>

Closing

We again thank the editor and reviewers for the detailed comments. The revised manuscript and Supplementary Information now make the data-generation process, optimizer formulation, comparison protocol, physical and climate scope, and remaining limitations explicit. We hope that the revised package addresses the reviewers' concerns and provides a clearer basis for evaluating surrogate-conditioned optimizer comparisons in block-scale urban energy design.